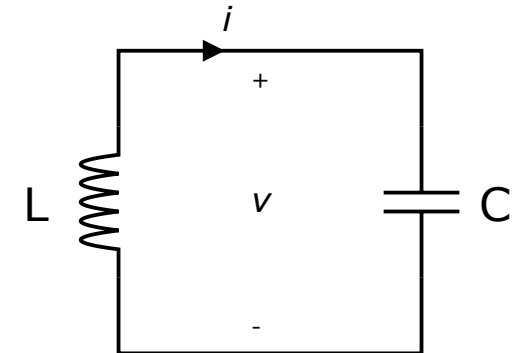
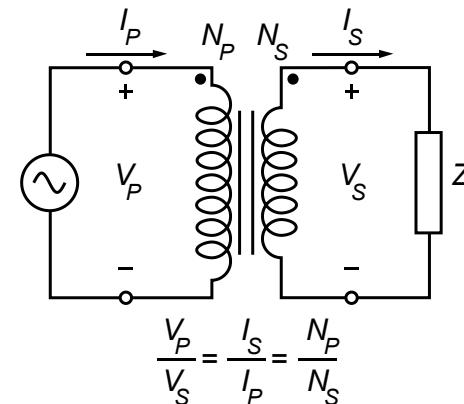
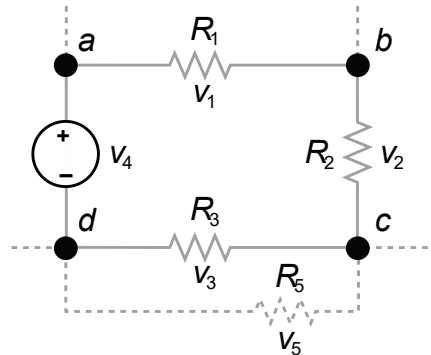
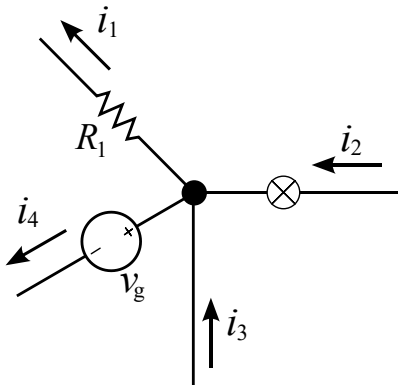


Introduction to Electrical Engineering



Overview

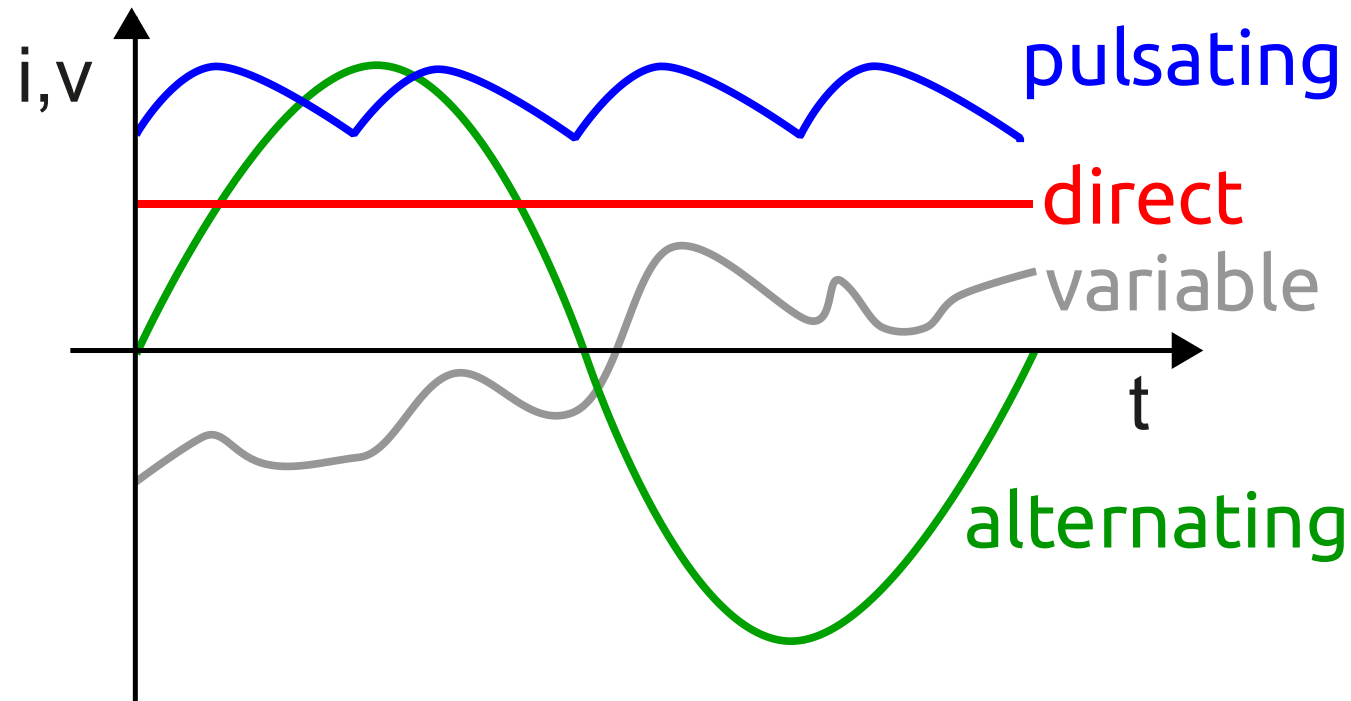
1. Physical Principles and Basic Concepts
2. Direct Current Circuits
3. Electric and Magnetic Fields
- 4. Alternating Current Technology**
5. Wrap-Up and Outlook

Alternating Current Technology

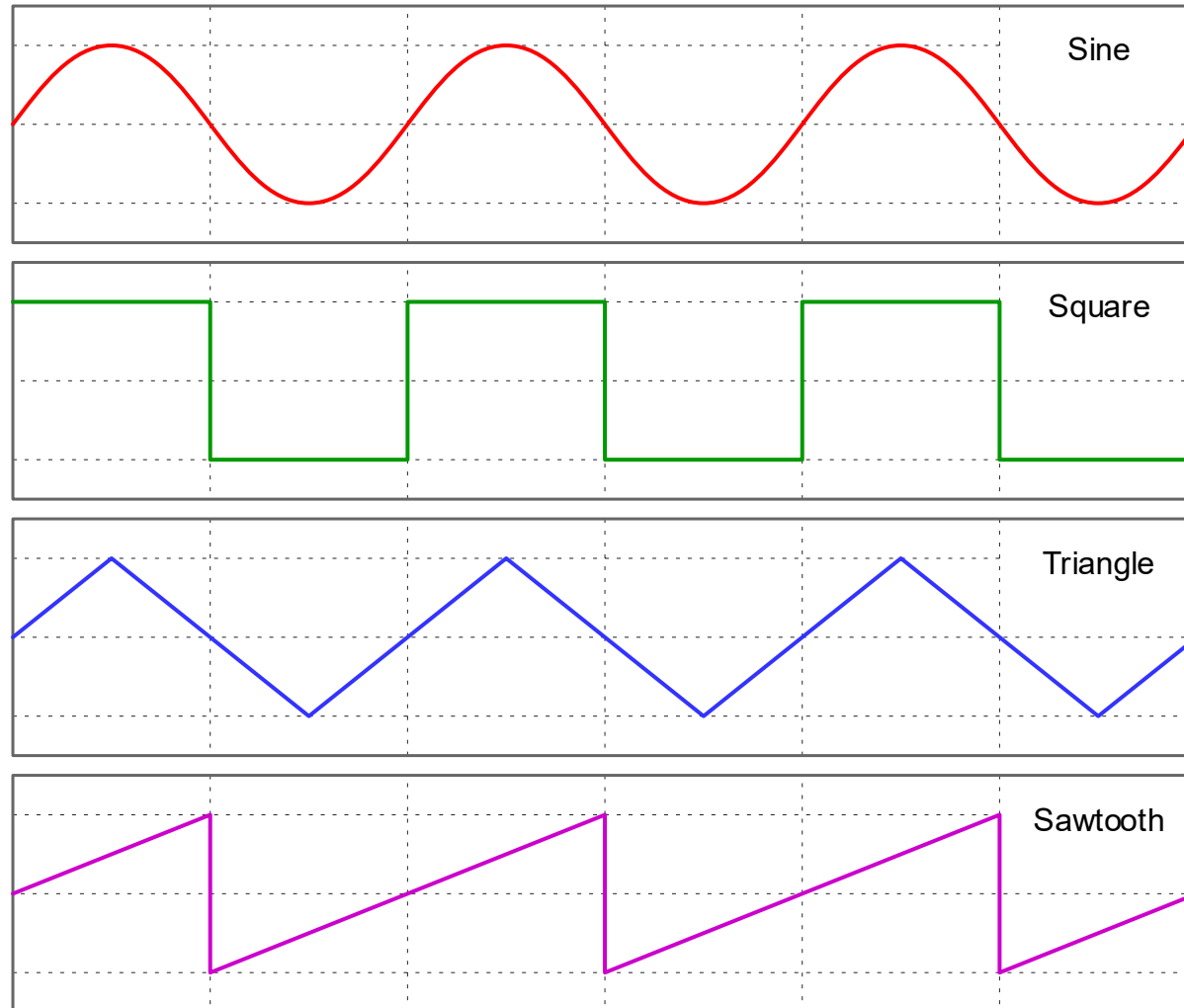
- Sinusoidal Waveform
- Phasor Representation
- Network Analysis in Phasor Domain
- AC Power
- High- & Low-Pass Filters
- Resonant Circuits
- Three-Phase Circuits

Alternating Current (AC)

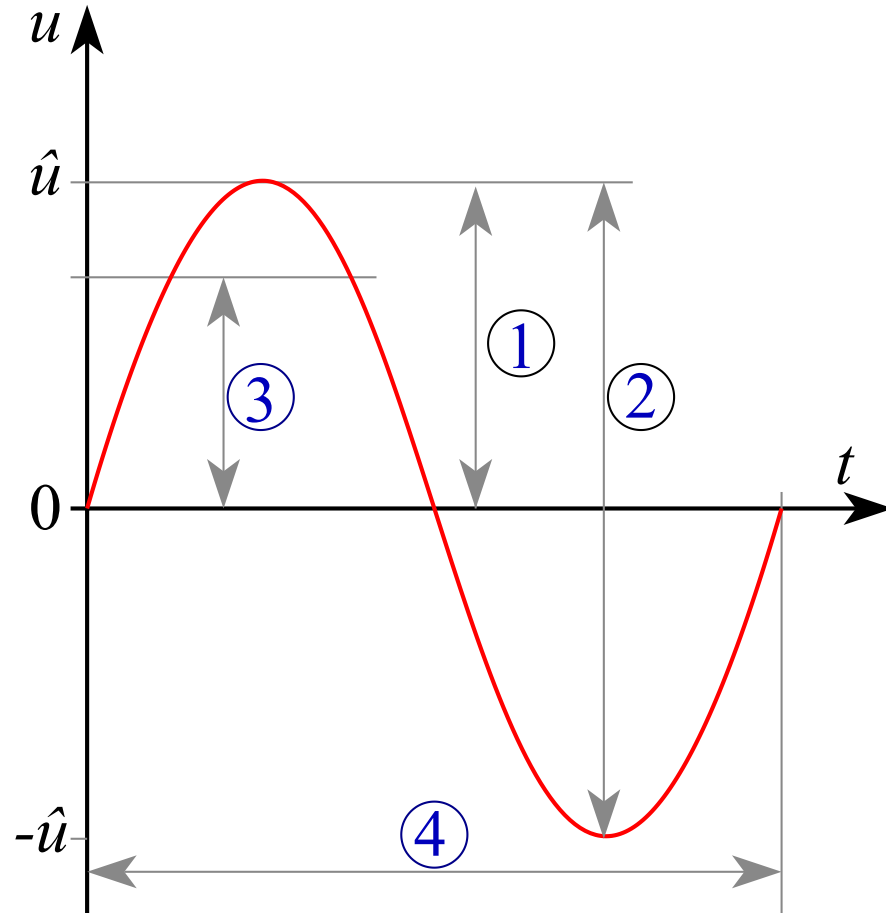
periodically reverses direction and changes its magnitude continuously with time



AC Waveforms



Sinusoidal AC



usual form of AC in most electric power circuits: sine wave

$$u(t) = \hat{u} \sin(\omega t), \quad i(t) = \hat{i} \sin(\omega t)$$

- $u(t) = u(t + nT)$

- frequency: $f = \frac{1}{T}$ $[f] = \frac{1}{s} = 1 \text{ Hz}$

- angular frequency: $\omega = 2\pi f$ $[\omega] = \frac{1}{s}$

1. peak voltage/current (amplitude) $\hat{u}, \hat{i}$
2. peak-to-peak amplitude: $2 \cdot \hat{u}$
3. RMS U, I
4. period (cycle time) T

Root Mean Square (RMS)

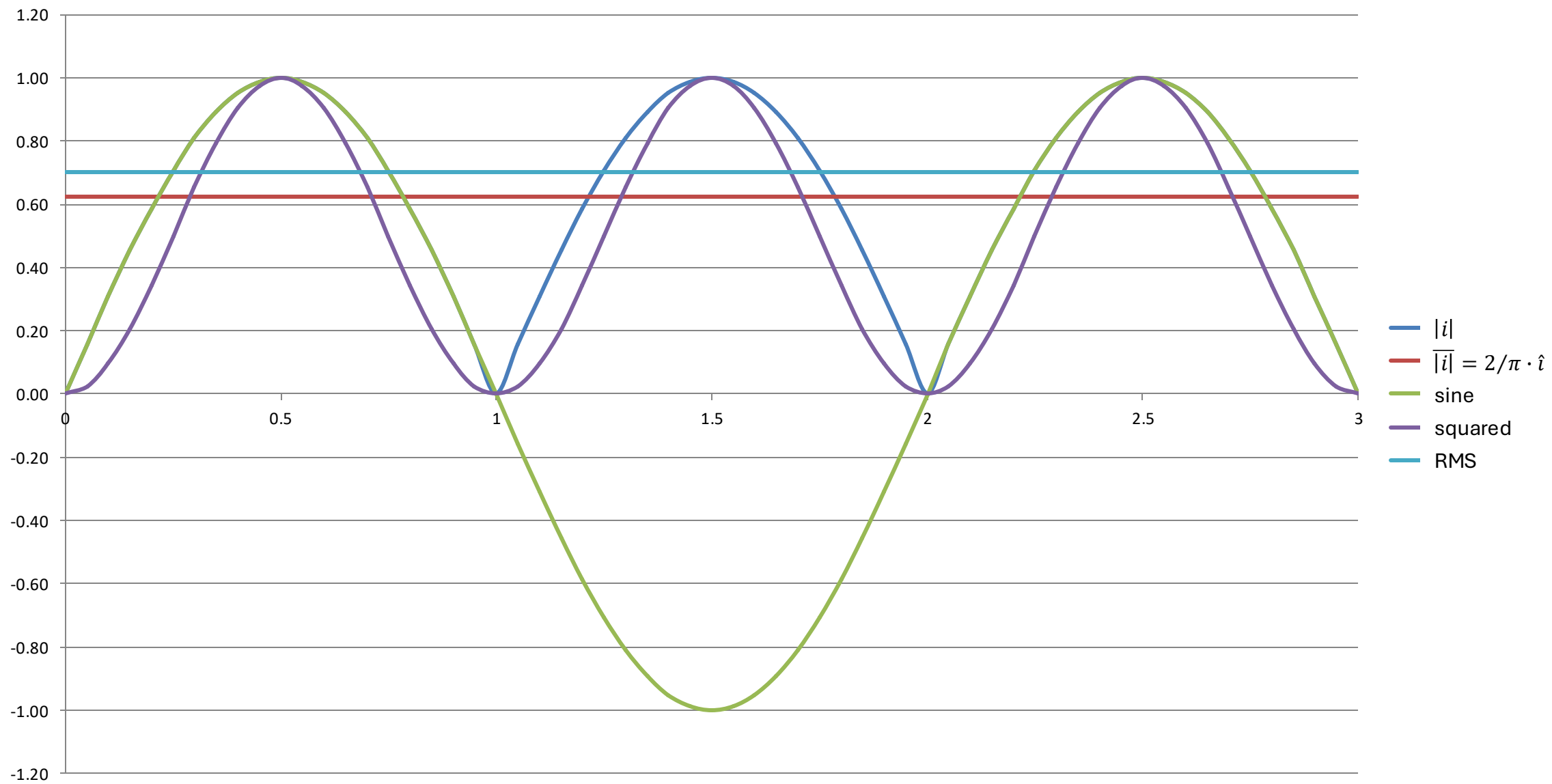
$$I = \sqrt{i^2} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} i^2(t) dt} = \frac{\hat{i}}{\sqrt{2}}$$

RMS of sum of mixed-frequency currents:
 $I_{\text{sum}} = \sqrt{I^2(f_1) + I^2(f_2)} \neq I(f_1) + I(f_2)$

The RMS of an AC corresponds to the DC value that generates the same amount of heat in a resistor. → effective value

mean itself is zero for pure AC:

$$\bar{i} = \frac{1}{T} \int_{t_0}^{t_0+T} i(t) dt = 0$$



Phase Shift

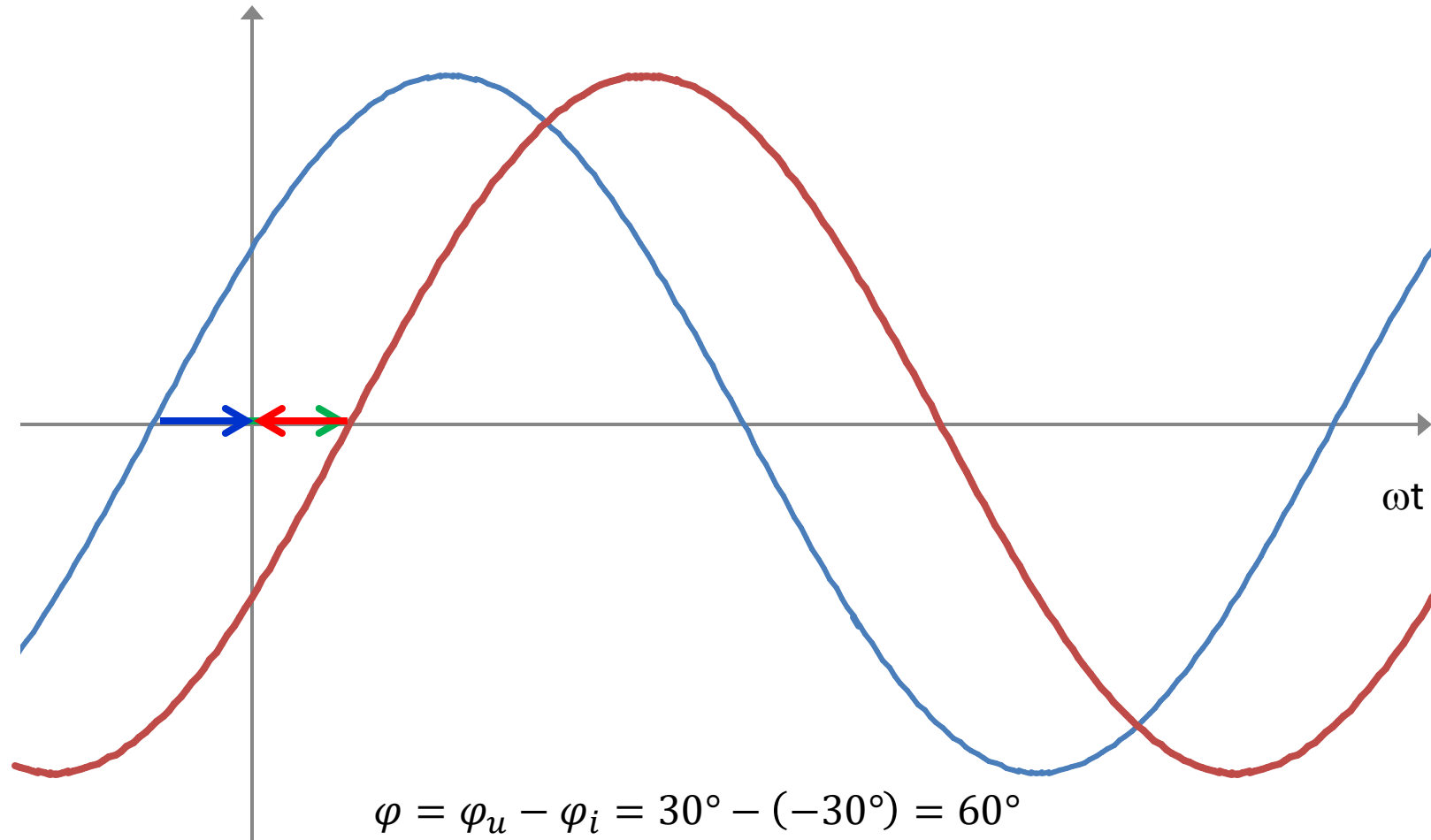
$$i(t) = \hat{i} \sin(\omega t + \varphi_0)$$

phase shift (for two currents with same frequency):

$$\varphi = \varphi_{01} - \varphi_{02}$$

phase shift between voltage and current (with current as reference):

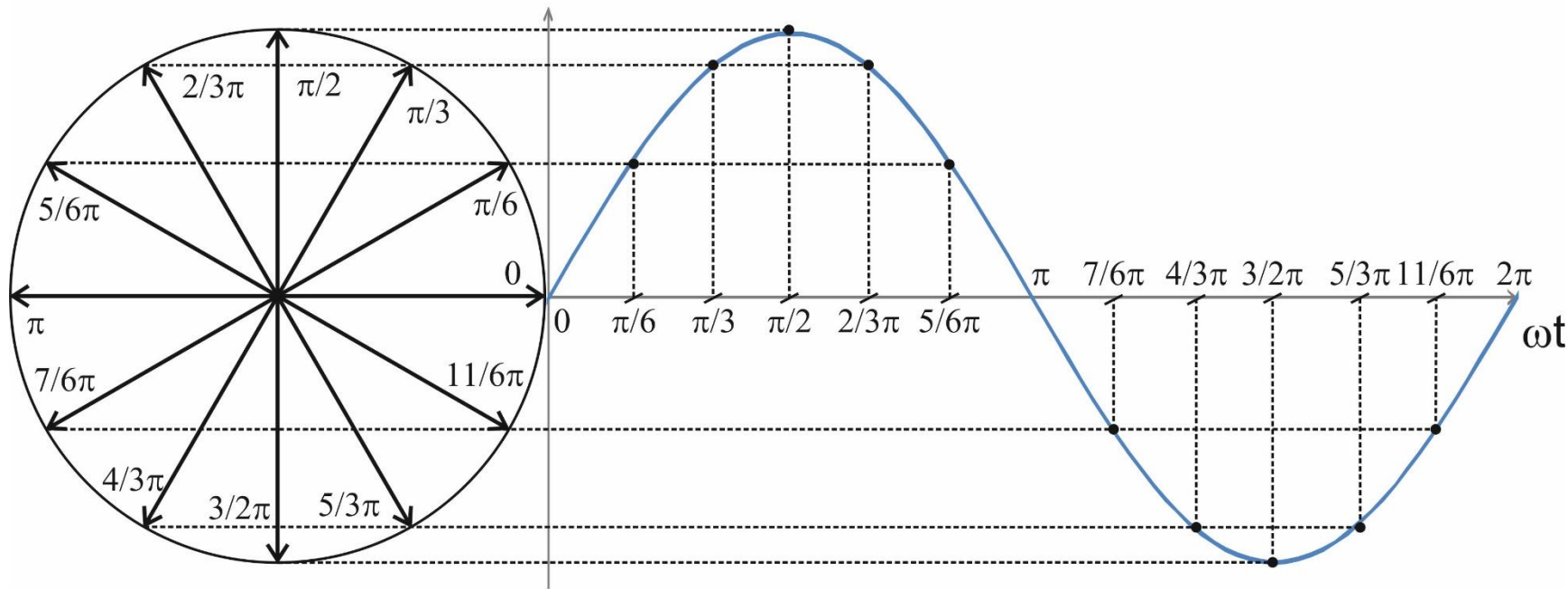
$$\varphi = \varphi_u - \varphi_i$$



In this example, the **current** lags behind the **voltage**.

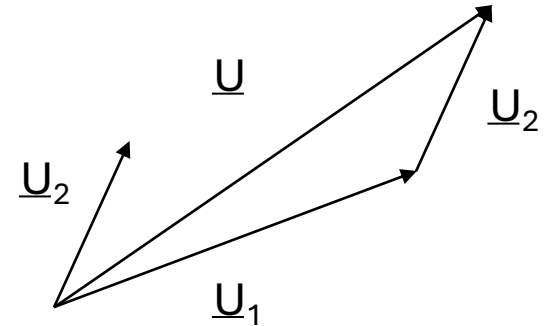
Phasor Representation

phasor diagram:

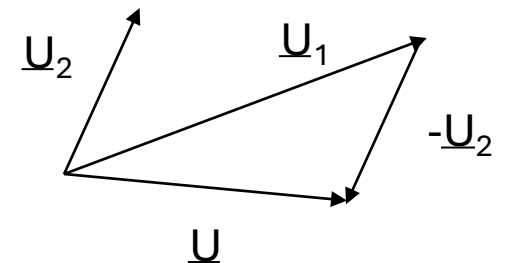


phasor calculations only work for quantities with same frequency
→ freeze rotation in time to get stationary phasors

addition:



subtraction:

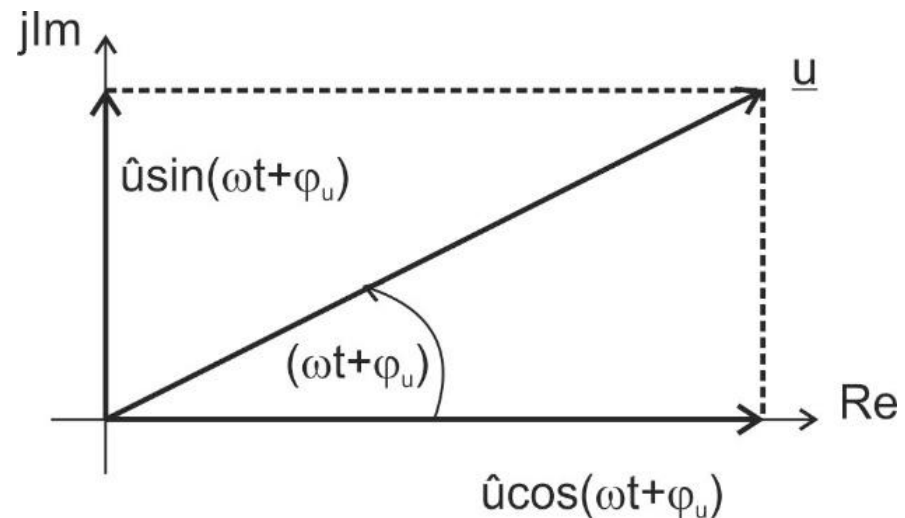


Use of Complex Numbers

$$u(t) = \hat{u} \cos(\omega t + \varphi_u) = \hat{u} \sin\left(\omega t + \varphi_u + \frac{\pi}{2}\right) = U \cdot \sqrt{2} \sin\left(\omega t + \varphi_u + \frac{\pi}{2}\right)$$

phasors can be represented by complex numbers:

$$\underline{u}(t) = \operatorname{Re}\{\underline{u}(t)\} + j \operatorname{Im}\{\underline{u}(t)\} = \hat{u} \cos(\omega t + \varphi_u) + j \hat{u} \sin(\omega t + \varphi_u)$$



Complex Amplitude

complex $\underline{u}(t) = \hat{u}[\cos(\omega t + \varphi_u) + j \sin(\omega t + \varphi_u)] = \hat{u} e^{j(\omega t + \varphi_u)} = \hat{u} e^{j\varphi_u} \underbrace{e^{j\omega t}}$

can be ignored if all quantities have the same frequency
→ stationary phasors

complex amplitude: $\underline{\hat{u}} = \hat{u} e^{j\varphi_u}$

$$\underline{u}(t) = \hat{u} e^{j\varphi_u} e^{j\omega t} = \underline{\hat{u}} e^{j\omega t}$$

complex RMS: $\underline{U} = U e^{j\varphi_u} = \frac{\underline{\hat{u}}}{\sqrt{2}}$

Transformation into Phasor Domain

time-dependent voltage	complex amplitude
$\hat{u} \cos(\omega t)$	$\underline{\hat{u}} = \hat{u}$
$\hat{u} \cos(\omega t + \varphi_u)$	$\underline{\hat{u}} = \hat{u} e^{j\varphi_u}$
$\hat{u} \sin(\omega t) = \hat{u} \cos\left(\omega t - \frac{\pi}{2}\right)$	$\underline{\hat{u}} = \hat{u} e^{-j\frac{\pi}{2}}$
$\hat{u} \sin(\omega t + \varphi_u) = \hat{u} \cos\left(\omega t + \varphi_u - \frac{\pi}{2}\right)$	$\underline{\hat{u}} = \hat{u} e^{j(\varphi_u - \frac{\pi}{2})}$

$$\underline{\hat{u}} = \hat{u}(\cos \varphi_u + j \sin \varphi_u) = Re + jIm$$

Back Transformation into Time Domain

1. multiplication with $e^{j\omega t}$ (phasors are stationary)
2. consider only real part of the complex number (or imaginary part if sine was used instead of cosine)

Derivation in Phasor Domain

$$\frac{d}{dt} \rightarrow j\omega$$

$$\left(\frac{d}{dt} e^{j\omega t} = e^{j\omega t} \cdot j\omega \right)$$

→ phase shift of $\frac{\pi}{2}$

$$\left(\underline{\hat{u}} \cdot j = \hat{u} e^{j\varphi_u} \cdot e^{j\frac{\pi}{2}} = \hat{u} e^{j(\varphi_u + \frac{\pi}{2})} \right)$$

example:

$$u(t) = L \frac{di}{dt} \rightarrow \underline{U} = j\omega L \underline{I}$$

Integration in Phasor Domain

$$\int dt \rightarrow \frac{1}{j\omega}$$

$$\left(\int e^{j\omega t} dt = e^{j\omega t} \cdot \frac{1}{j\omega} \right)$$

→ phase shift of $-\frac{\pi}{2}$

$$\left(\underline{\hat{u}}/j = \hat{u} e^{j\varphi_u} \cdot e^{-j\frac{\pi}{2}} = \hat{u} e^{j(\varphi_u - \frac{\pi}{2})} \right)$$

example:

$$i(t) = \frac{1}{L} \int u(t) dt \rightarrow \underline{I} = \frac{1}{j\omega L} \underline{U}$$

Complex Resistance

impedance:

$$\frac{\underline{u}(t)}{\underline{i}(t)} = \frac{\hat{\underline{u}} e^{j\omega t}}{\hat{\underline{i}} e^{j\omega t}} = \frac{|\hat{\underline{u}}| e^{j\varphi_u}}{|\hat{\underline{i}}| e^{j\varphi_i}} = |\underline{Z}| e^{j(\varphi_u - \varphi_i)} = |\underline{Z}| e^{j\varphi} = \underline{Z} \quad \text{abstract quantity}$$

Ohm's law in phasor domain: $\underline{\hat{u}} = \underline{Z} \underline{\hat{i}}$ or $\underline{U} = \underline{Z} \underline{I}$

$$\underline{Z} = R + jX = |\underline{Z}| e^{j\varphi}$$

resistance

reactance

$$|\underline{Z}| = \sqrt{R^2 + X^2}$$
$$\tan \varphi = \frac{X}{R}$$

impedance angle

modulus of impedance
(apparent impedance)

Complex Conductance

admittance:

abstract quantity

$$\underline{Y} = \frac{1}{\underline{Z}} = \frac{1}{R + jX} = \underbrace{\frac{R}{R^2 + X^2}}_{\text{conductance } G} + j \underbrace{\frac{-X}{R^2 + X^2}}_{\text{susceptance } B}$$

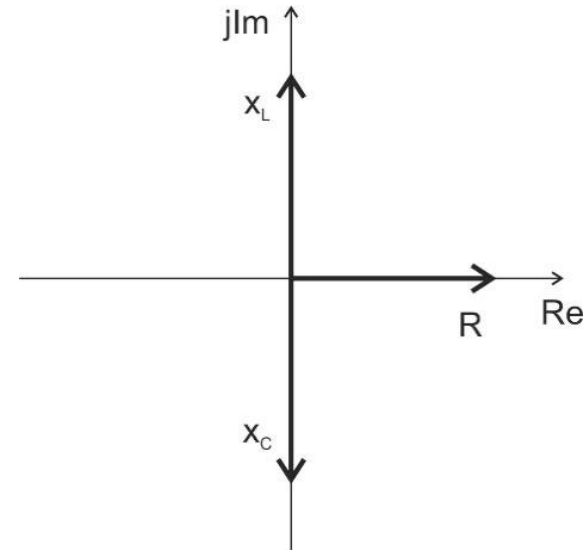
$$\underline{Y} = \frac{1}{\underline{Z}} = \frac{1}{|\underline{Z}|e^{j\varphi}} = |\underline{Y}|e^{-j\varphi}$$

admittance angle
 $\theta = -\varphi$

modulus of admittance
(apparent admittance)

Linear Passive Components

R	$\underline{\hat{u}} = R \cdot \underline{\hat{i}}$	$\underline{Z} = R$
L	$\underline{\hat{u}} = j\omega L \cdot \underline{\hat{i}}$	$\underline{Z}_L = j\omega L = jX_L$
C	$\underline{\hat{u}} = \frac{1}{j\omega C} \cdot \underline{\hat{i}}$	$\underline{Z}_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C} = jX_C$

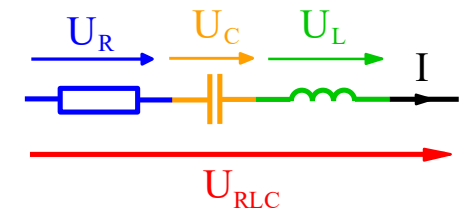
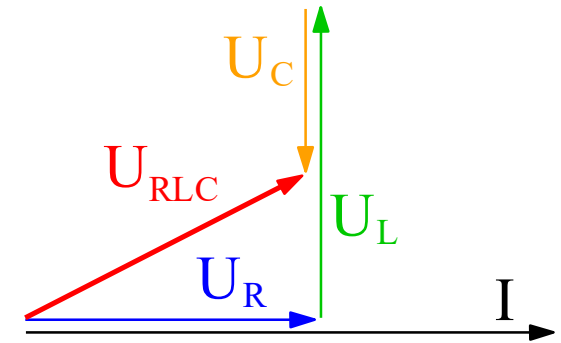


R: U and I in phase ($\varphi = 0$)

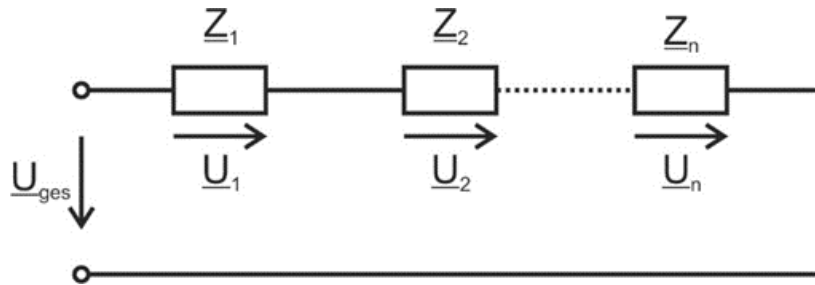
L: U leads I by 90°

C: U lags I by 90°

must consider angles:

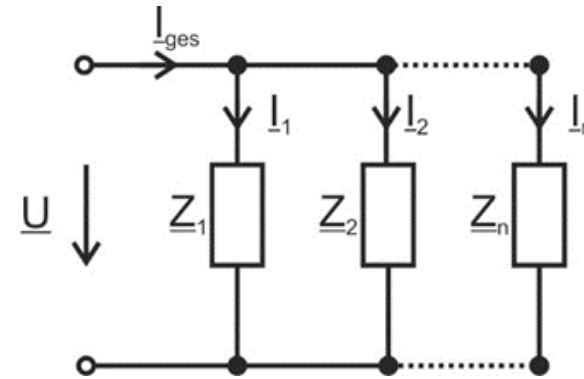


Network Analysis in Phasor Domain



$$\underline{Z} = \sum_{k=1}^n \underline{Z}_k$$

$$\frac{\underline{U}_1}{\underline{U}_2} = \frac{\underline{Z}_1}{\underline{Z}_2}$$



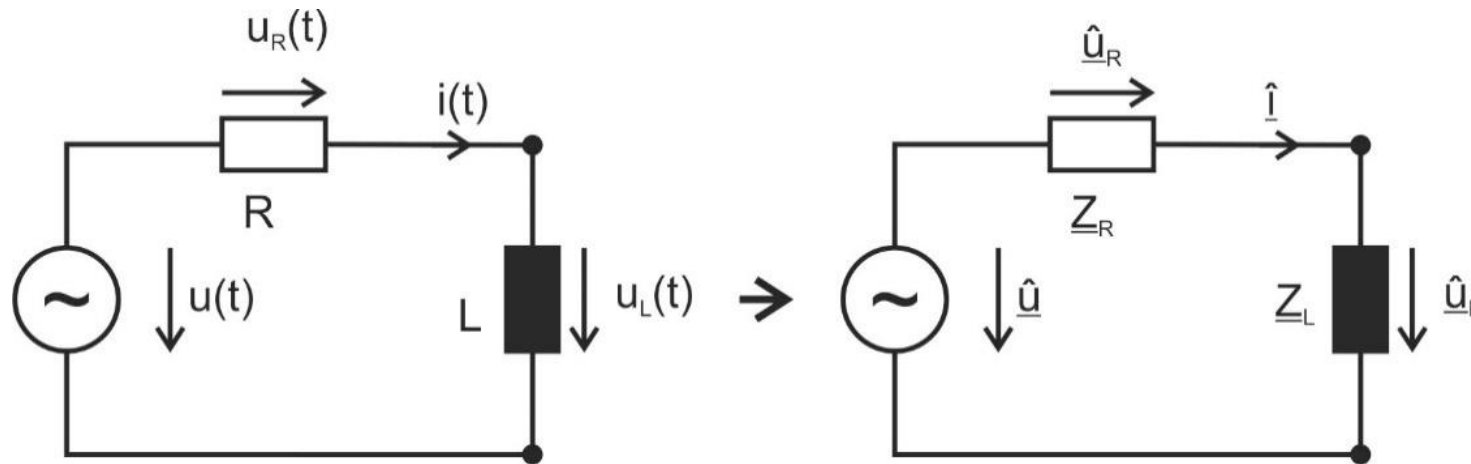
$$\frac{1}{\underline{Z}} = \sum_{k=1}^n \frac{1}{\underline{Z}_k}$$

$$\frac{\underline{I}_1}{\underline{I}_2} = \frac{\underline{Z}_2}{\underline{Z}_1}$$

KCL and KVL also valid for complex amplitudes and RMS values

Example 1

The current through the circuit and the voltage across the coil are to be determined.



$$u(t) = \hat{u} \cos(\omega t + \varphi_u) \rightarrow \underline{\hat{u}} = \hat{u} e^{j\varphi_u}$$

Example 1

$$\underline{\hat{u}} = \underline{\hat{u}}_R + \underline{\hat{u}}_L = \underline{Z}_R \cdot \underline{\hat{i}} + \underline{Z}_L \cdot \underline{\hat{i}} = (R + j\omega L)\underline{\hat{i}}$$

$$\underline{\hat{i}} = \frac{\underline{\hat{u}}}{R + j\omega L} = \frac{\underline{\hat{u}} \cdot (R - j\omega L)}{R^2 + \omega^2 L^2} = \underline{\hat{u}} \frac{R}{R^2 + \omega^2 L^2} - j\underline{\hat{u}} \frac{\omega L}{R^2 + \omega^2 L^2}$$

$$\hat{i} = \frac{\hat{u}}{Z} = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}}$$

$$\varphi = \tan^{-1} \frac{\omega L}{R} = \varphi_u - \varphi_i$$

$$\left(\varphi_u = \tan^{-1} \frac{\omega L}{R}; \quad \varphi_i = 0 \quad \text{or} \quad \varphi_u = 0; \quad \varphi_i = \tan^{-1} \frac{-\omega L}{R} \right)$$

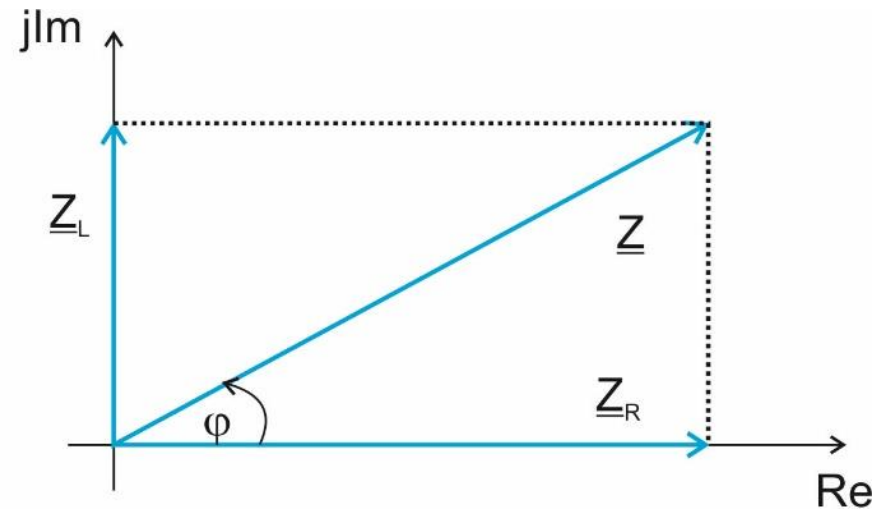
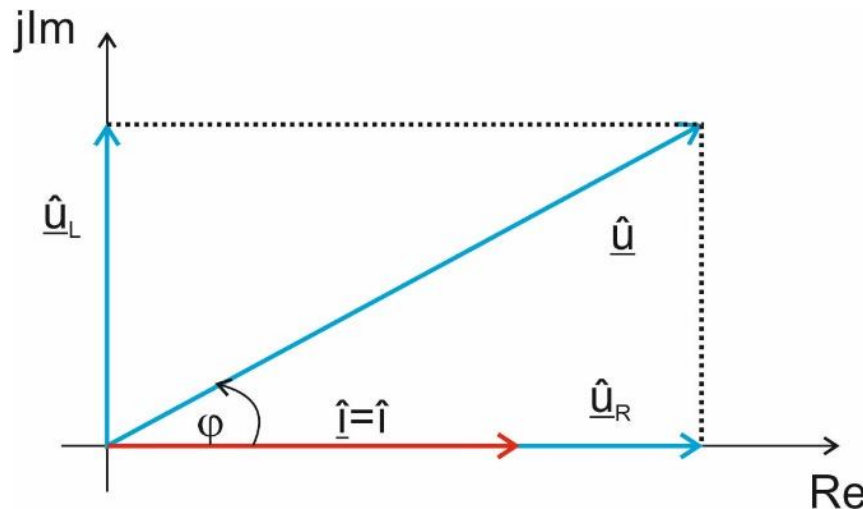
$$\underline{\hat{i}} = \hat{i} \cdot e^{j\varphi_i} = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}} e^{j(\varphi_u - \varphi)}$$

Example 1

$$i(t) = \text{Re}\{\underline{\hat{i}} \cdot e^{j\omega t}\} = \text{Re}\{\hat{i} \cdot e^{j\varphi_i} e^{j\omega t}\} = \hat{i} \cos(\omega t + \varphi_i) = \frac{\hat{u}}{\sqrt{R^2 + (\omega L)^2}} \cos\left(\omega t + \varphi_u - \tan^{-1} \frac{\omega L}{R}\right)$$

$$\underline{\hat{u}}_L = \underline{Z}_L \cdot \underline{\hat{i}} = j\omega L \underline{\hat{i}} = e^{j\frac{\pi}{2}} \omega L \hat{i} e^{j\varphi_i} = \hat{i} \omega L e^{j(\varphi_i + \frac{\pi}{2})}$$

$$u_L(t) = \text{Re}\{\underline{\hat{u}}_L e^{j\omega t}\} = \text{Re}\{\hat{i} \omega L e^{j(\omega t + \varphi_i + \frac{\pi}{2})}\} = \hat{u} \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}} \cos\left(\omega t + \frac{\pi}{2} + \varphi_u - \tan^{-1} \frac{\omega L}{R}\right)$$



Example 1 with Numbers

$$\underline{Z} = \underline{Z}_R + \underline{Z}_L = R + j\omega L = (50 + j 31.42) \Omega$$

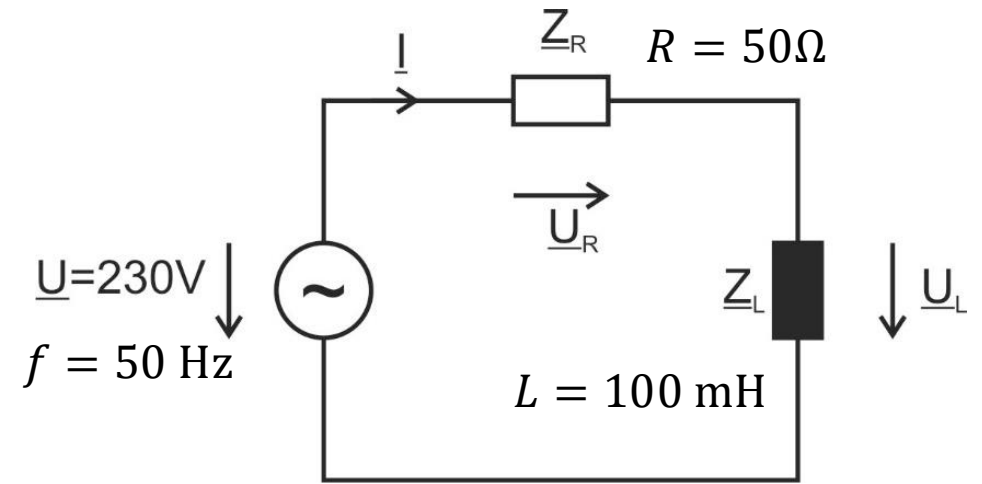
$$Z = \sqrt{R^2 + (\omega L)^2} = 59.1 \Omega$$

$$\varphi = \tan^{-1} \frac{\omega L}{R} = 32.1^\circ$$

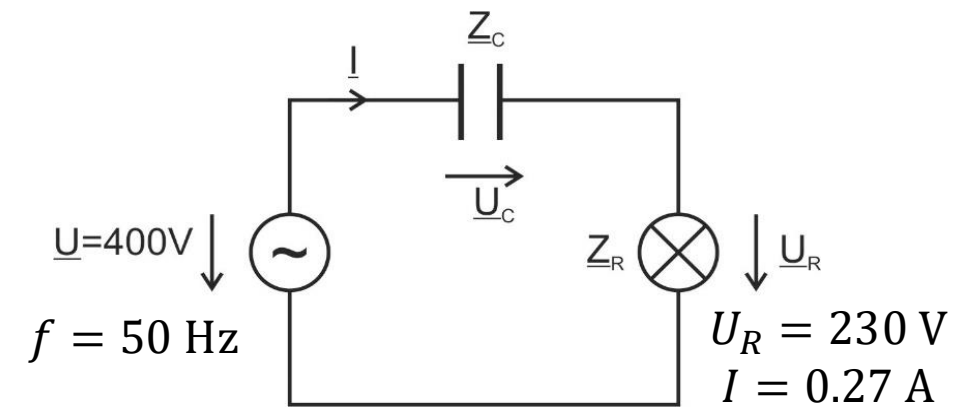
$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = \frac{U}{(R + j\omega L)} = \frac{U(R - j\omega L)}{(R^2 + (\omega L)^2)} = (3.30 - j 2.07) \text{ A} = 3.90 \text{ A} \cdot e^{-j 32.1^\circ}$$

$$\underline{U}_R = \underline{Z}_R \cdot \underline{I} = 50 \Omega \cdot 3.90 \text{ A} \cdot e^{-j 32.1^\circ} = 195 \text{ V} \cdot e^{-j 32.1^\circ} = (165.19 - j 103.63) \text{ V}$$

$$\underline{U}_L = \underline{Z}_L \cdot \underline{I} = j\omega L \cdot \underline{I} = j 31.42 \Omega \cdot (3.30 - j 2.07) \text{ A} = (65.04 + j 103.69) \text{ V} = 122.40 \text{ V} \cdot e^{j 57.9^\circ}$$



Example 2



What capacitance must the capacitor have so that the light bulb is connected to the voltage $U_R = 230\text{ V}$?

$$U^2 = U_C^2 + U_R^2 \rightarrow U_C = \sqrt{400^2 - 230^2}\text{ V} = 327.26\text{ V}$$

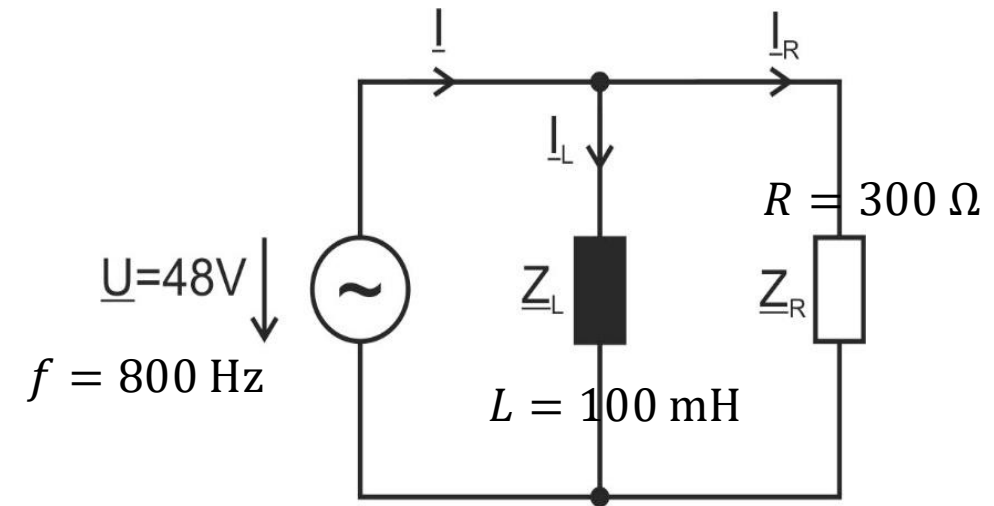
$$\frac{1}{\omega C} = \frac{U_C}{I} \rightarrow C = 2.63\text{ }\mu\text{F}$$

Example 3

$$\frac{1}{\underline{Z}} = \frac{1}{R} + \frac{1}{j\omega L} = \frac{j\omega L + R}{j\omega RL}$$

$$\rightarrow \underline{Z} = \frac{\omega^2 RL^2 + j\omega R^2 L}{R^2 + (\omega L)^2} = (221.2 + j 132.0) \Omega$$

$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = (160 - j 95) \text{ mA}$$



Instantaneous Power

$$p(t) = u(t) \cdot i(t)$$

For a purely ohmic resistance, current and voltage are in phase:

$$\begin{aligned} p(t) &= \hat{u} \sin \omega t \cdot \hat{i} \sin \omega t = \hat{u} \hat{i} \sin^2 \omega t \\ &= \hat{u} \hat{i} \frac{1}{2} (1 - \cos 2\omega t) = UI (1 - \cos 2\omega t) \end{aligned}$$

$p(t) > 0$ for passive sign convention $\rightarrow$ passive component

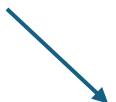
Average Power

area under power curve: electrical energy converted into heat in a resistor

average power (aka active power for resistance):

$$\bar{P} = \frac{1}{T} \int_0^T p(t) dt$$

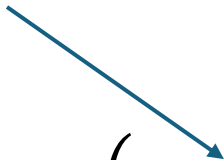
resistance


$$= \frac{1}{T} \int_0^T \frac{1}{2} \hat{u} \hat{i} (1 - \cos 2\omega t) dt = \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(t - \frac{1}{2\omega} \sin 2\omega t \right) \Bigg|_0^T$$

$$= \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(T - \frac{1}{2\omega} \sin 2\omega T \right) = \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(T - \frac{1}{2\omega} \sin 2\omega \frac{2\pi}{\omega} \right) = \frac{1}{2} \hat{u} \hat{i} = UI$$

Reactive Power

for reactance (capacitor or inductor): phase shift of $\frac{\pi}{2}$ between $u(t)$ and $i(t)$


$$p(t) = \hat{u} \sin \omega t \cdot \hat{i} \sin \left(\omega t + \frac{\pi}{2} \right) = \hat{u} \hat{i} \sin \omega t \cos \omega t = \hat{u} \hat{i} \frac{1}{2} \sin 2\omega t = UI \sin 2\omega t$$

$$\bar{P} = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{T} \int_0^T \frac{1}{2} \hat{u} \hat{i} (\sin 2\omega t) dt = \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(-\frac{1}{2\omega} \cos 2\omega t \right) \Bigg|_0^T$$

$$= \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(-\frac{1}{2\omega} (\cos 2\omega T + 1) \right) = \frac{1}{2} \hat{u} \hat{i} \frac{1}{T} \left(-\frac{1}{2\omega} (\cos 4\pi + 1) \right) = 0$$

→ not converted into heat or useful work

Apparent Power

mixed case for resistive-capacitive or resistive-inductive load:

$$p(t) = \sqrt{2} U \sin \omega t \cdot \sqrt{2} I \sin(\omega t - \varphi) = \dots = UI \cdot \cos \varphi \cdot (1 - \cos 2\omega t) - UI \cdot \sin \varphi \sin 2\omega t$$

- purely active for $\varphi = 0$
- purely reactive for $\varphi = \frac{\pi}{2}$ or $\varphi = -\frac{\pi}{2}$

average power: $P = UI \cdot \cos \varphi$

apparent power: $S = U \cdot I$
 $[S] = 1 \text{ VA}$

electrical devices must be designed for the apparent power to be transmitted

Complex Power

$$\underline{S} = \underline{U} \cdot \underline{I}^* = U e^{j\varphi_u} I e^{-j\varphi_i} = S e^{j(\overbrace{\varphi_u - \varphi_i}^{\varphi})}$$

$$\underline{S} = \underline{U} \left(\frac{\underline{U}}{\underline{Z}} \right)^* = \underline{U} \cdot \underline{U}^* \frac{1}{\underline{Z}^*} = \frac{U^2}{\underline{Z}^*}$$

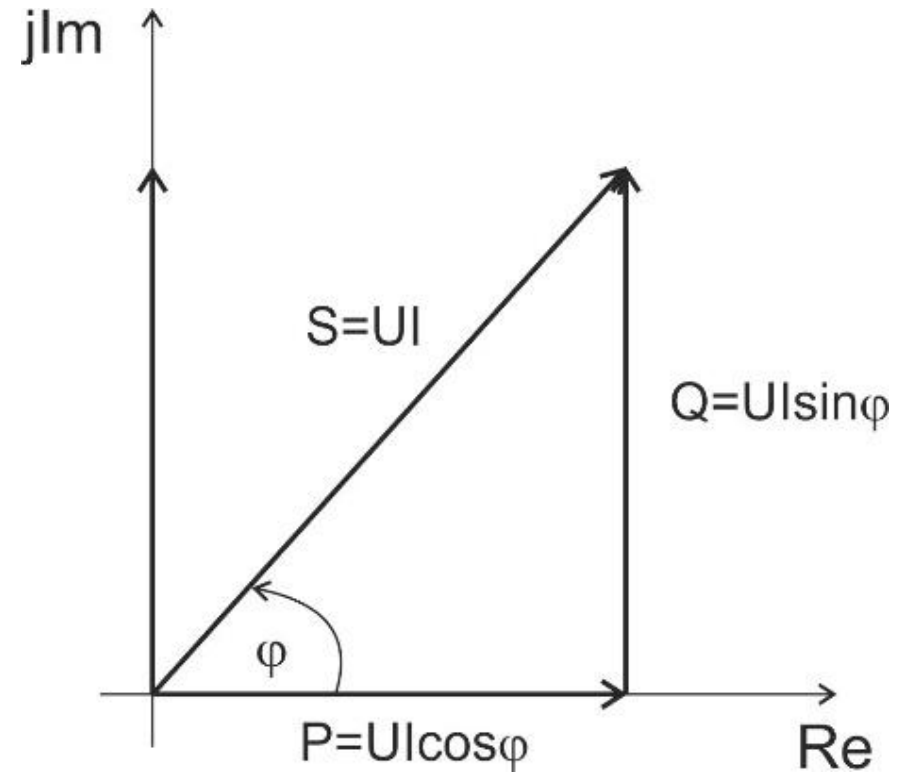
$$\underline{S} = \frac{1}{2} \hat{\underline{u}} \cdot \hat{\underline{i}}^* = P + jQ$$

active power

reactive power
[Q] = 1 VAR

with $S = \sqrt{P^2 + Q^2} = U \cdot I$

apparent power



$$\varphi = \tan^{-1} \frac{Q}{P}; \quad \cos \varphi = \frac{P}{S}; \quad \sin \varphi = \frac{Q}{S}$$

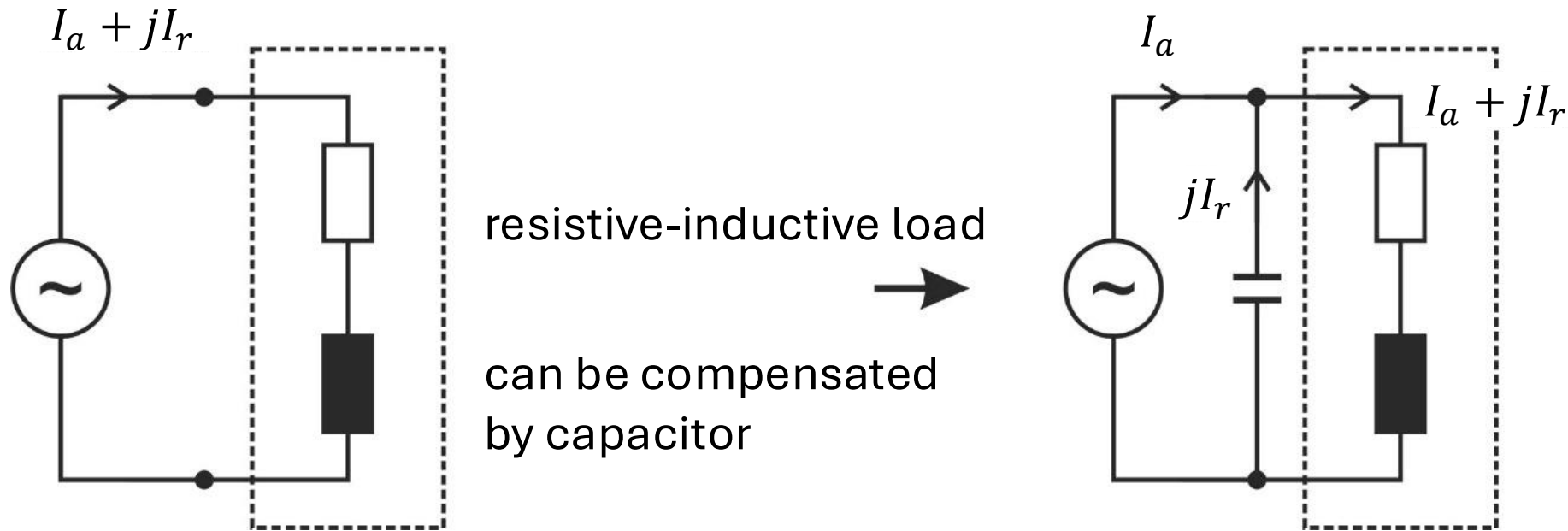
power factor:

measure of how effectively electrical power is being converted into useful work output

Reactive Power Compensation

The absorption of inductive reactive power corresponds to the output of capacitive reactive power and vice versa: $\varphi = 180^\circ$

→ can compensate each other (useful to avoid transmission losses)



Example

A motor ($U = 230 \text{ V}$, $f = 50 \text{ Hz}$, $I = 16 \text{ A}$, $\cos \varphi = 0.8$) is to be compensated using a capacitor.

without capacitor:

$$I_a = I \cdot \cos \varphi = 12.8 \text{ A}, \quad I_r = I \cdot \sin \varphi = -9.6 \text{ A} \rightarrow \underline{I} = (12.8 - j 9.6) \text{ A}$$

$$\underline{S} = \underline{U} \cdot \underline{I}^* = 230 \text{ V} \cdot (12.8 + j 9.6) \text{ A}$$

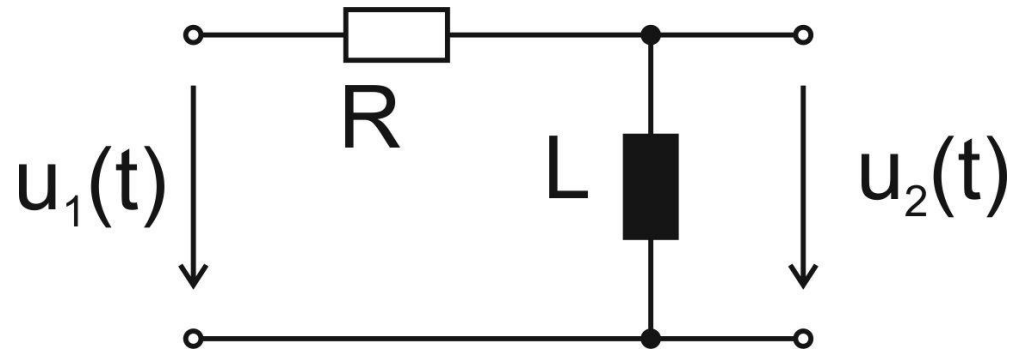
$$S = \sqrt{2.944^2 + 2.208^2} \text{ kVA} = 3.68 \text{ kVA}, \quad \varphi = \tan^{-1} \frac{2.208}{2.944} = 36.87^\circ$$

reactive current needs to be compensated by capacitor:

$$\frac{1}{\omega C} = \frac{U}{I_r} \rightarrow C = 133 \text{ } \mu\text{F}$$

$$\underline{S} = \underline{U} \cdot \underline{I}^* = 230 \text{ V} \cdot 12.8 \text{ A} = 2.944 \text{ kVA} = S, \quad \varphi = 0$$

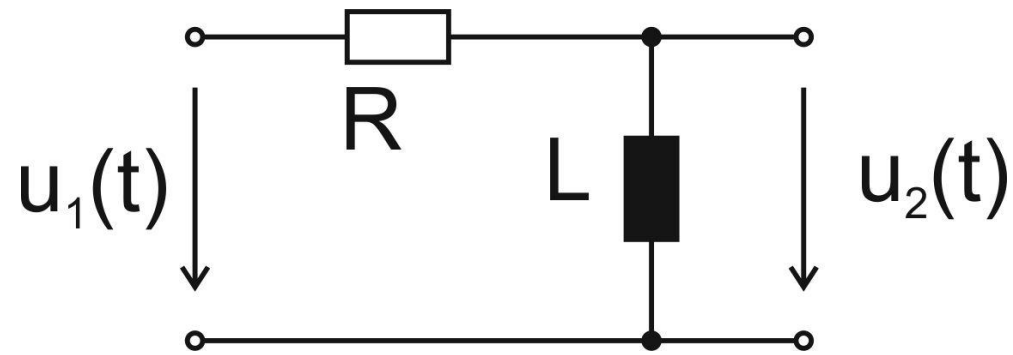
Frequency-Dependent Voltage Divider



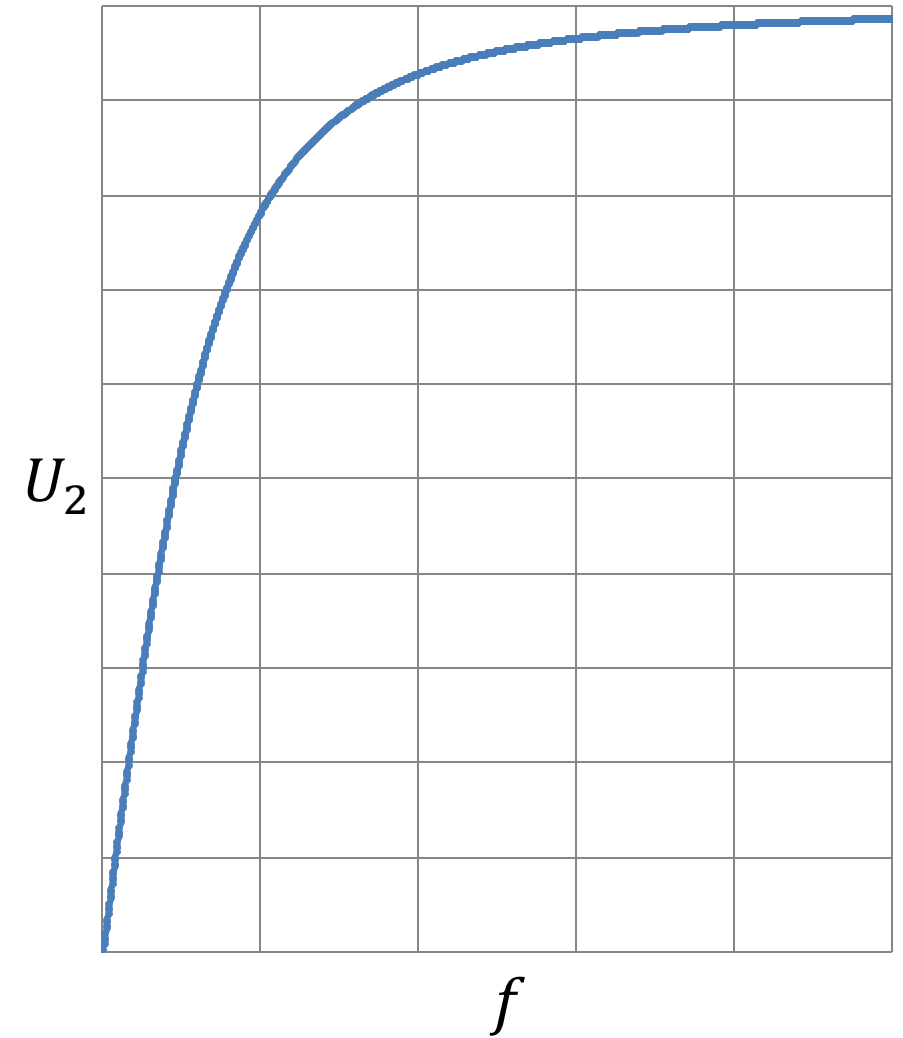
$$\underline{U}_2 = \underline{U}_1 \cdot \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} = \underline{U}_1 \cdot \frac{j\omega L}{R + j\omega L} = \underline{U}_1 \cdot \frac{(\omega L)^2 - j\omega LR}{R^2 + (\omega L)^2}$$

$$\begin{aligned} U_2 &= U_1 \cdot \sqrt{\left(\frac{(\omega L)^2}{R^2 + (\omega L)^2}\right)^2 + \left(\frac{\omega LR}{R^2 + (\omega L)^2}\right)^2} = U_1 \cdot \sqrt{\frac{(\omega L)^4 + (\omega LR)^2}{(R^2 + (\omega L)^2)^2}} \\ &= U_1 \frac{\sqrt{(\omega L)^2((\omega L)^2 + R^2)}}{R^2 + (\omega L)^2} = U_1 \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}} = U_1 \cdot \frac{Z_1}{Z} \end{aligned}$$

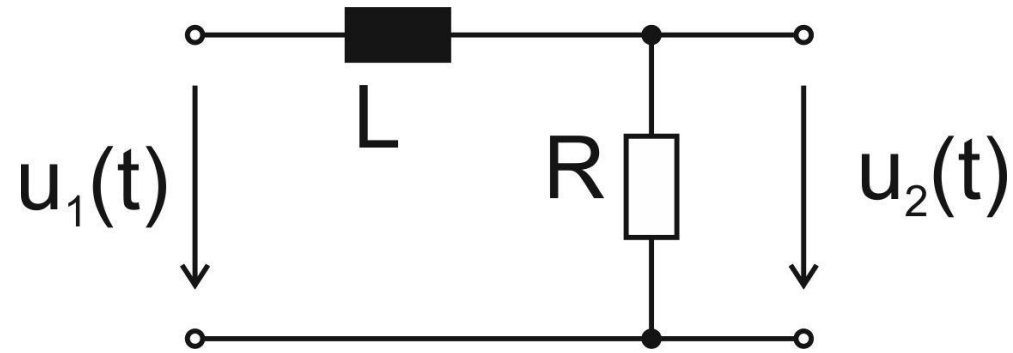
High-Pass RL Filter



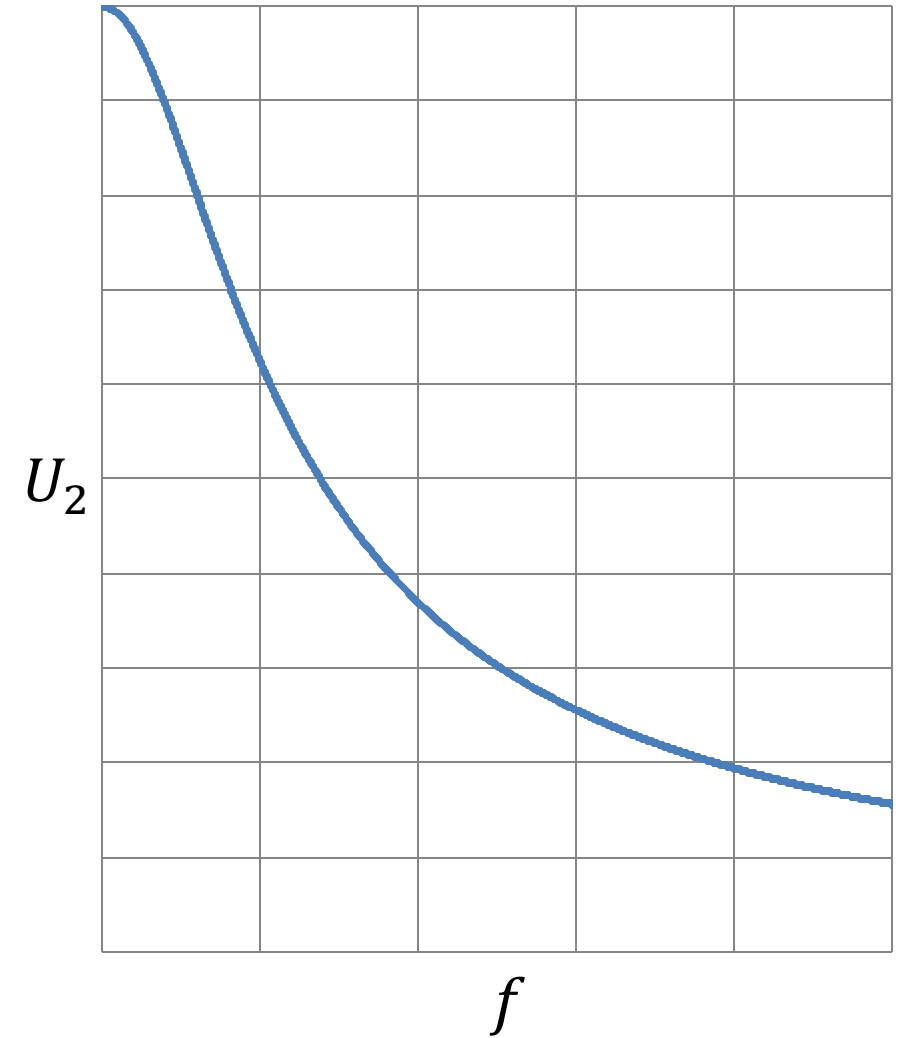
$$U_2 = U_1 \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}}$$



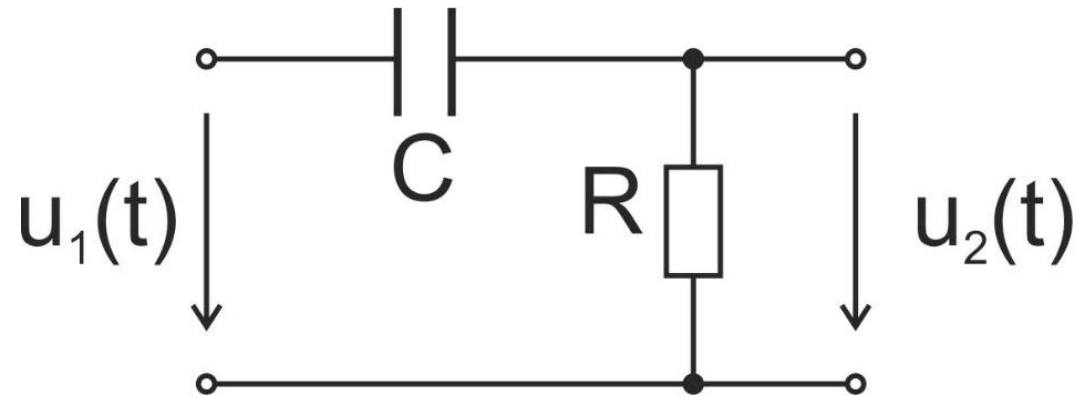
Low-Pass RL Filter



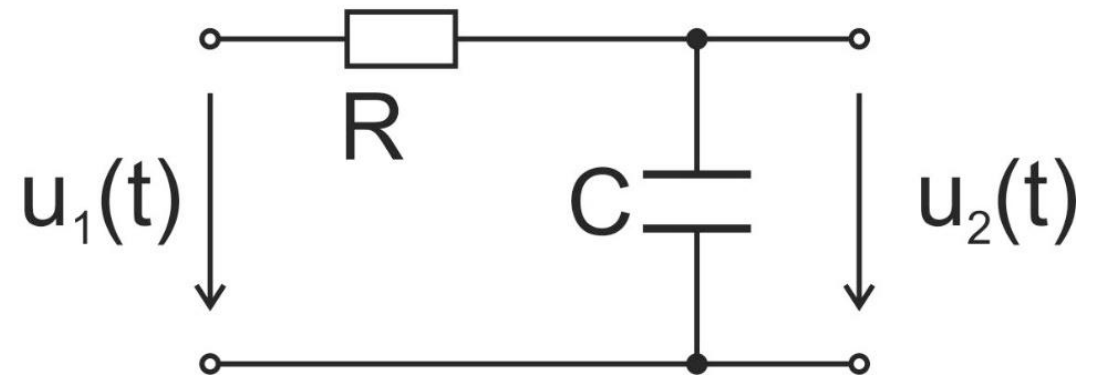
$$U_2 = U_1 \frac{R}{\sqrt{R^2 + (\omega L)^2}}$$



RC Filters



high-pass filter



low-pass filter

first-order filters: either a capacitor or an inductor

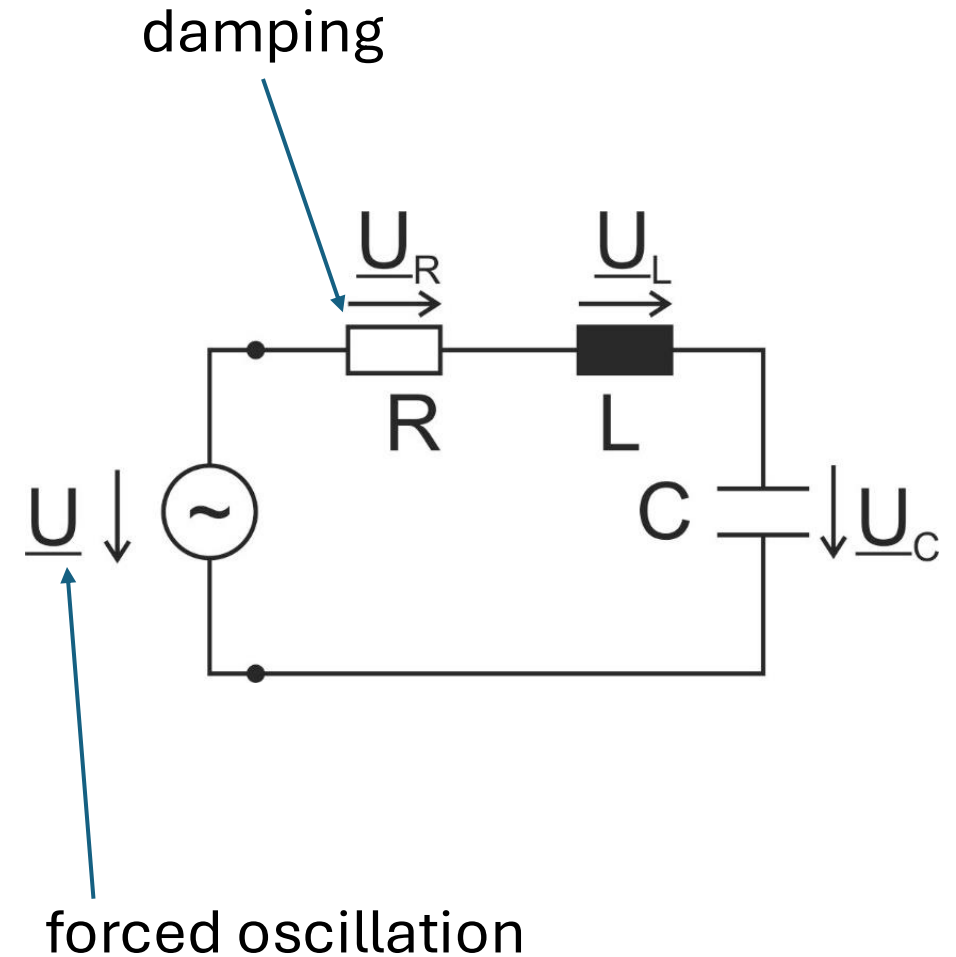
Series Resonant Circuit

$$\underline{Z} = R + j \left(\omega L - \frac{1}{\omega C} \right)$$

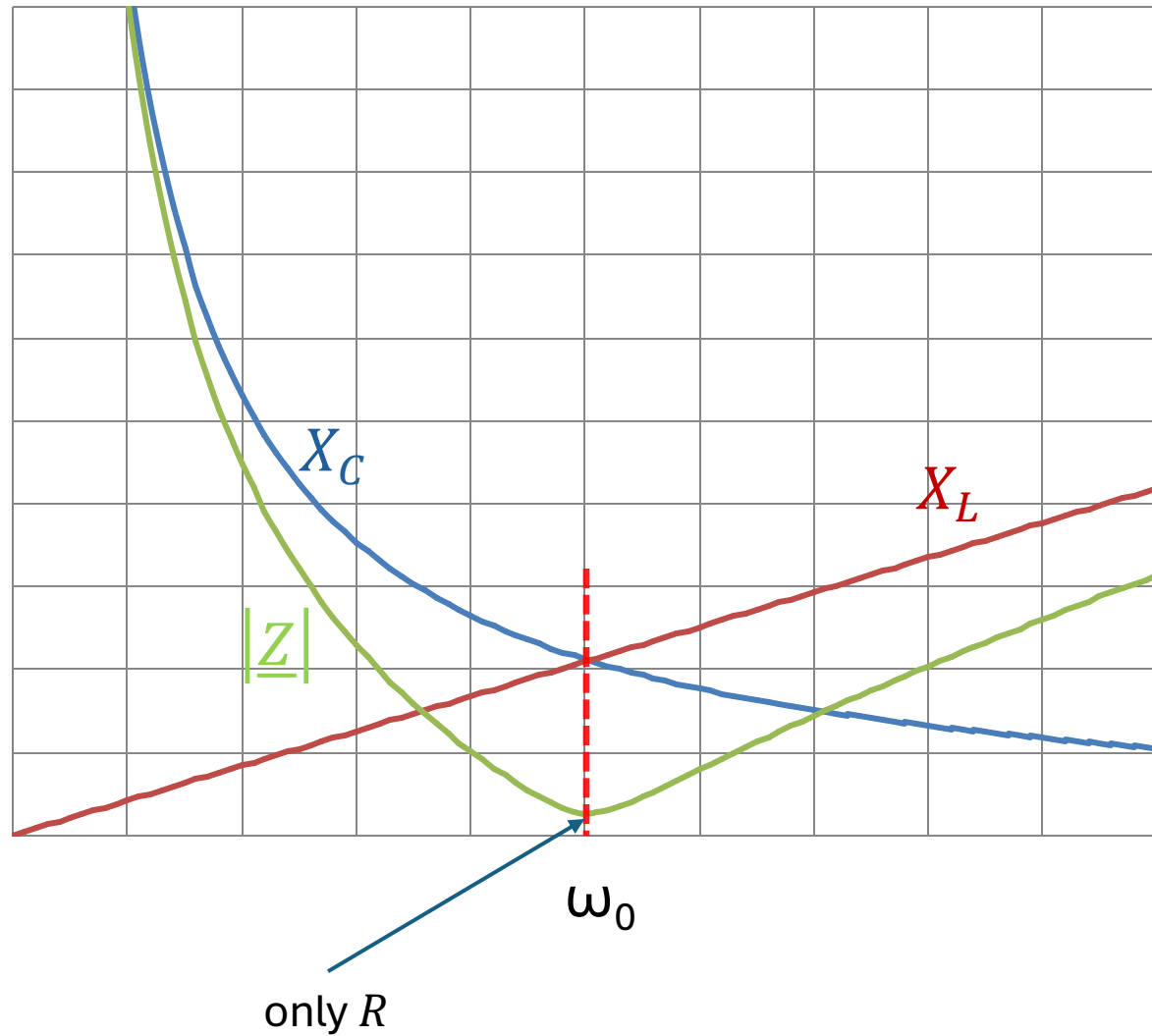
resonance frequency f_0 : inductive and capacitive reactances cancel each other out

$$\omega_0 L - \frac{1}{\omega_0 C} = 0$$
$$\rightarrow \omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

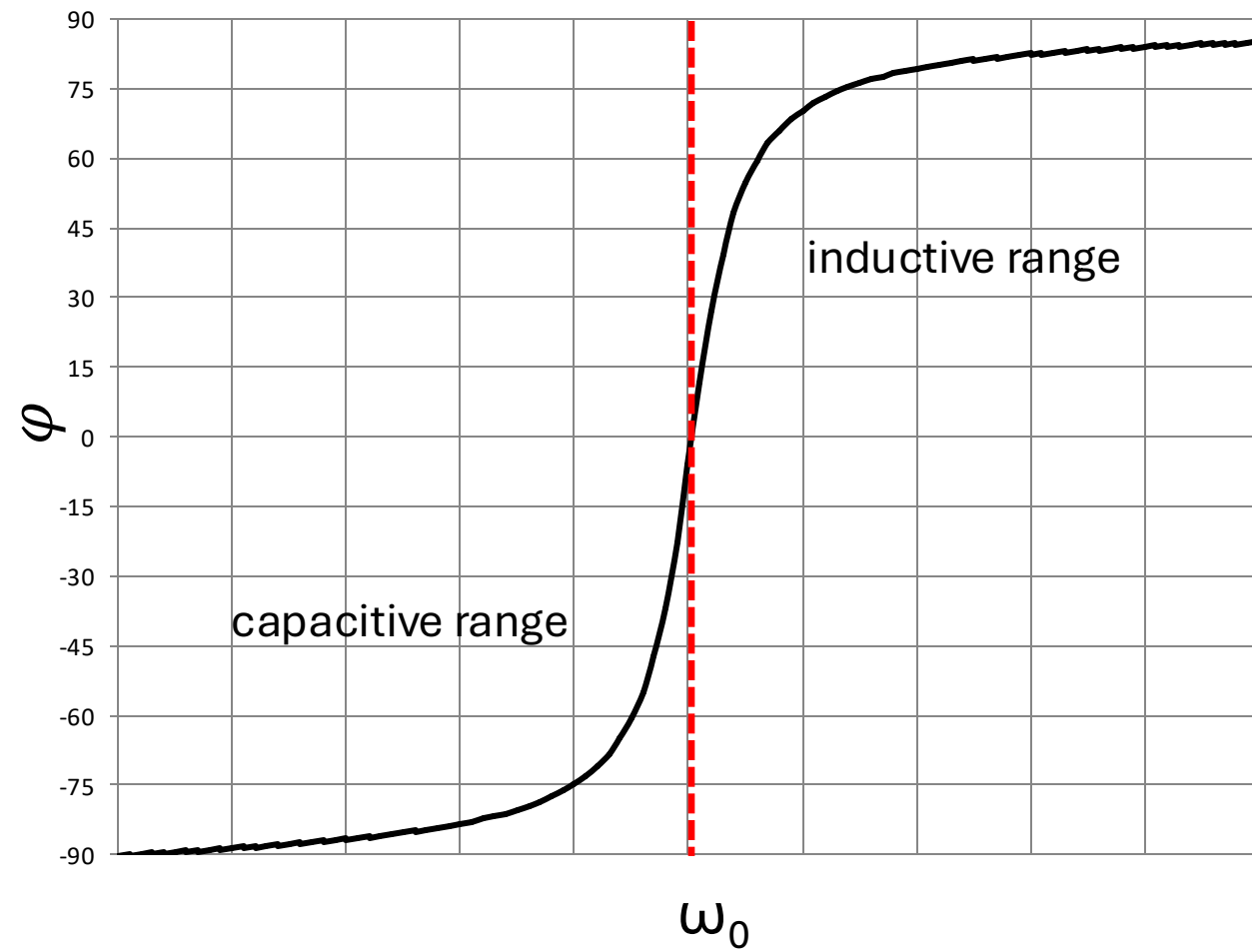
$$\underline{Z}(\omega_0) = R$$



impedance minimal at resonance:



phase shift between source voltage and current:



Voltage Resonance

$$U_L = U \frac{\omega L}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$U_C = U \frac{1}{\omega C \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

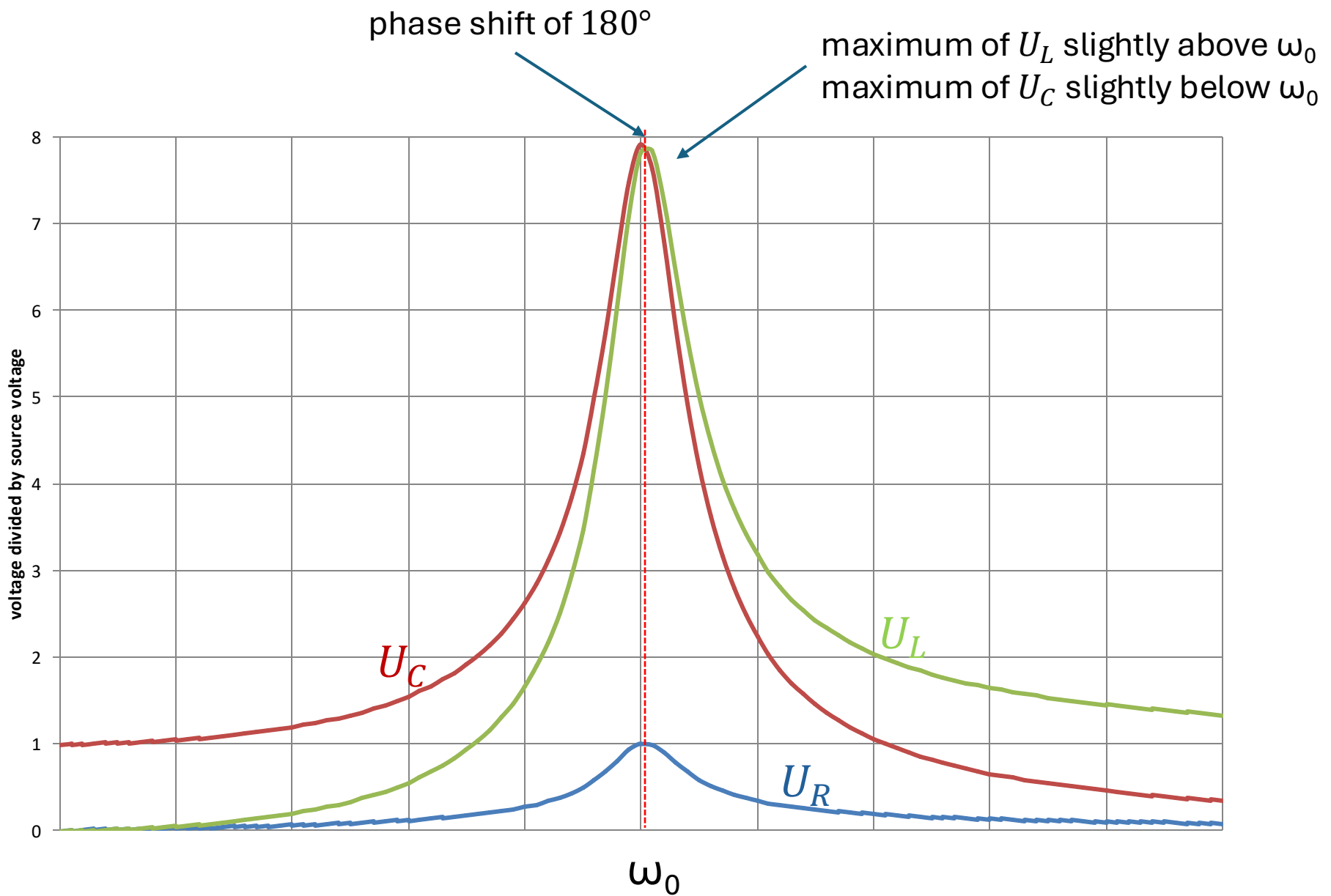
for ω_0 : $U_L = U_C = U \frac{1}{R} \sqrt{\frac{L}{C}} = U \cdot Q$

$$\underline{U}_L = -\underline{U}_C$$

quality factor $Q = \frac{1}{R} \sqrt{\frac{L}{C}}$

can be > 1

→ voltages across inductor and capacitor can be significantly larger than the source voltage (resonance catastrophe)



$$U_R = U \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \rightarrow \text{maximum for } \omega = \omega_0: U_R = U \quad (\text{also maximum current})$$

Voltage Maxima

$$\frac{dU_L}{d\omega_L} = 0 \quad \rightarrow \quad \omega_L = \sqrt{\frac{2}{2LC - R^2 C^2}} = \sqrt{\frac{2}{2LC - \frac{LC}{Q^2}}} = \omega_0 \frac{1}{\sqrt{1 - \frac{1}{2Q^2}}}$$

$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$ $\omega_0 = \frac{1}{\sqrt{LC}}$

$$\frac{dU_C}{d\omega_C} = 0 \quad \rightarrow \quad \omega_C = \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}} = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$$

→ no maxima for $Q \leq \frac{1}{\sqrt{2}}$

resonance behavior more pronounced the higher Q

Parallel Resonant Circuit

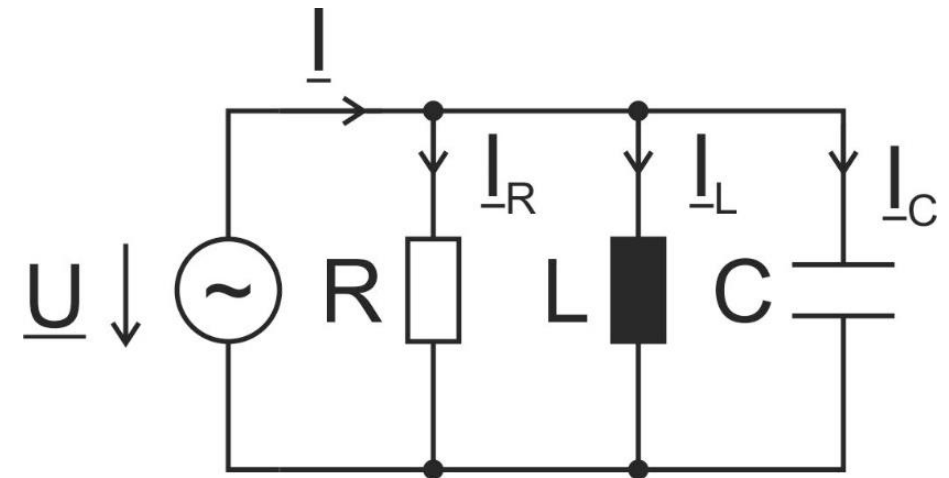
$$\underline{Y} = G + j \left(\omega C - \frac{1}{\omega L} \right)$$

$$\omega_0 C - \frac{1}{\omega_0 L} = 0$$
$$\rightarrow \omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$\underline{Y}(\omega_0) = G$$

admittance minimal at resonance (impedance maximal) $\rightarrow$ total current minimal ($\underline{I}_L = -\underline{I}_C$)

below ω_0 : determined by inductance
above ω_0 : determined by capacitance



Current Resonance

$$I_R = I \frac{G}{\sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

$$I_L = I \frac{1}{\omega L \sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

$$I_C = I \frac{\omega C}{\sqrt{G^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

resonance:

$$I_R = I = \frac{U}{R}$$

$$I_L = I_C = IR \sqrt{\frac{C}{L}} = I \cdot Q$$

→ current through inductor and capacitor can be larger than the source current

Current Maxima

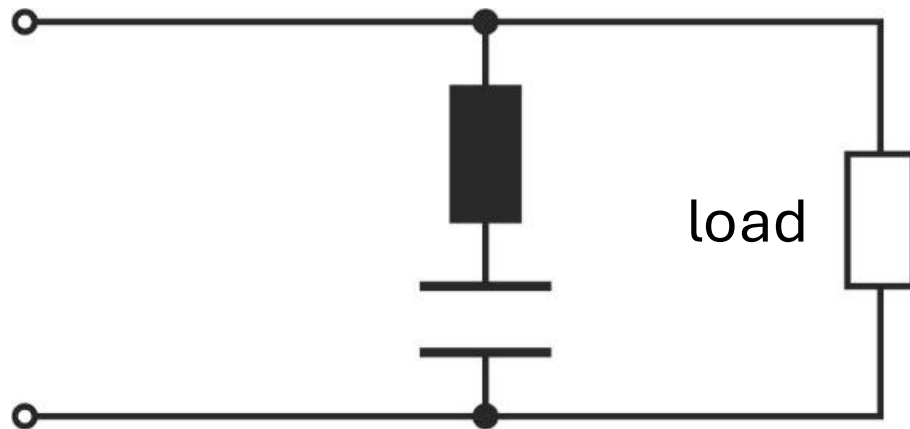
$$\frac{dI_L}{d\omega_L} = 0 \quad \rightarrow \quad \omega_L = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$$

$$\frac{dI_C}{d\omega_C} = 0 \quad \rightarrow \quad \omega_C = \omega_0 \frac{1}{\sqrt{1 - \frac{1}{2Q^2}}}$$

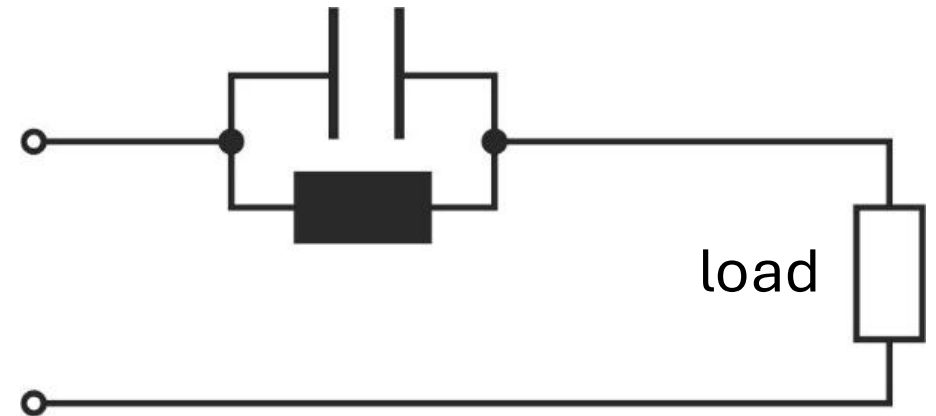
$\rightarrow$ again, no maxima for $Q \leq \frac{1}{\sqrt{2}}$

Second-Order Filters

acceptor circuit:
admittance peak at resonance



rejector circuit:
impedance peak at resonance



Three-Phase Circuits

idea:

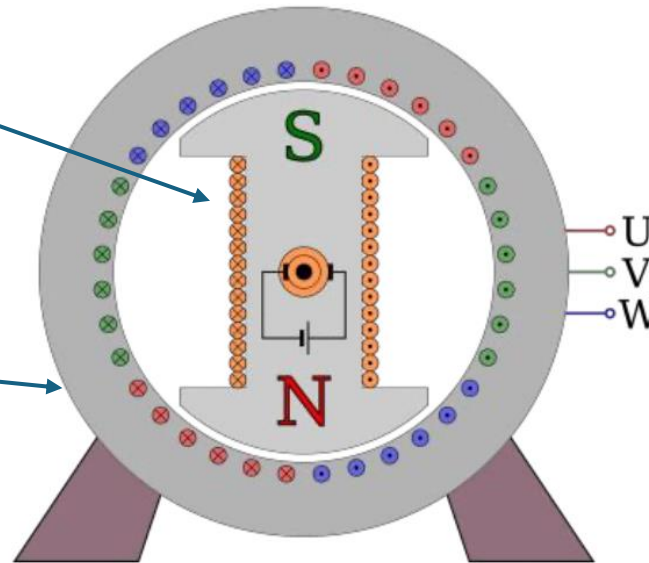
three alternating currents of the same frequency and amplitude but phase-shifted by 120° from each other

motivation:

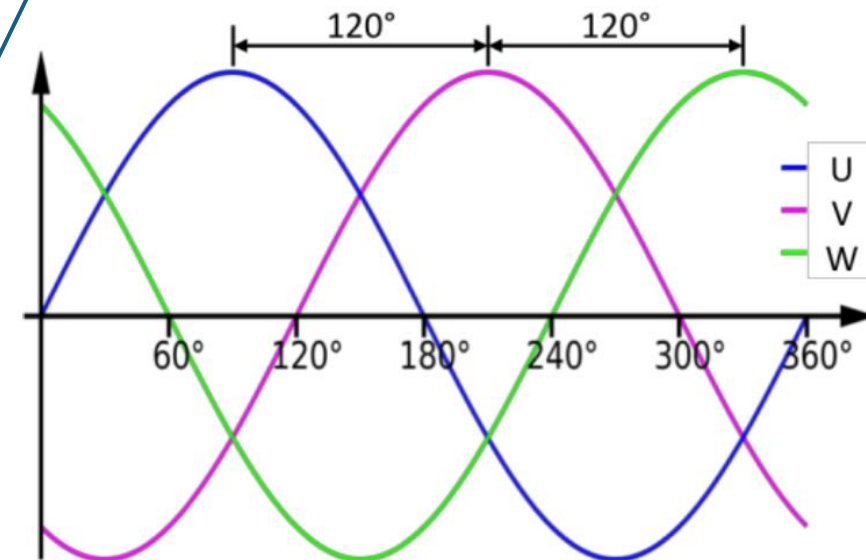
- constant power transmission
- generation of rotating magnetic field → electric motor

Three-Phase Generator

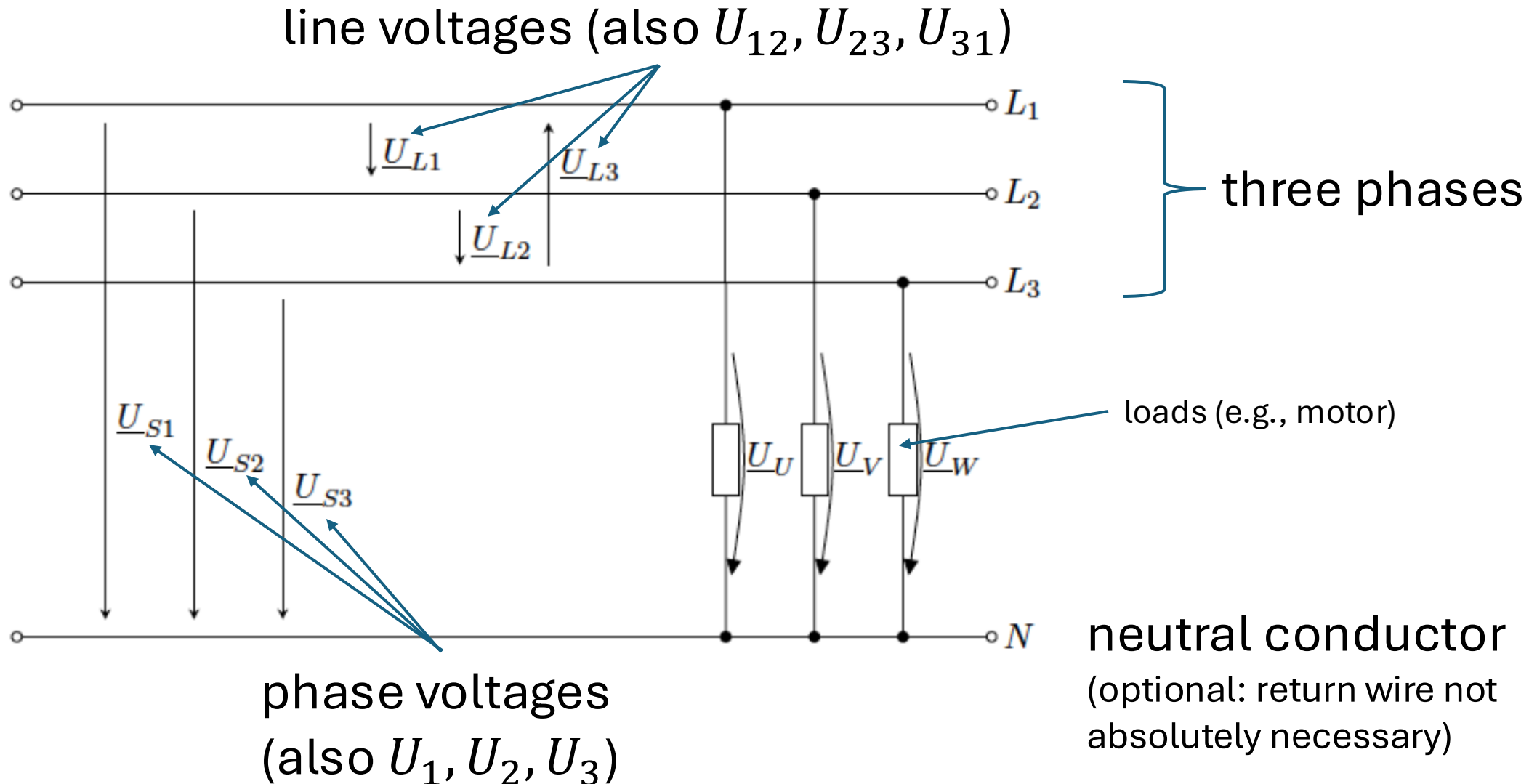
- rotor: (mechanically) rotating magnetic field (generated by coil with DC or permanent magnet)
- stator: three coils offset by 120°
- induces sinusoidal alternating voltage in each coil, phase-shifted by 120°



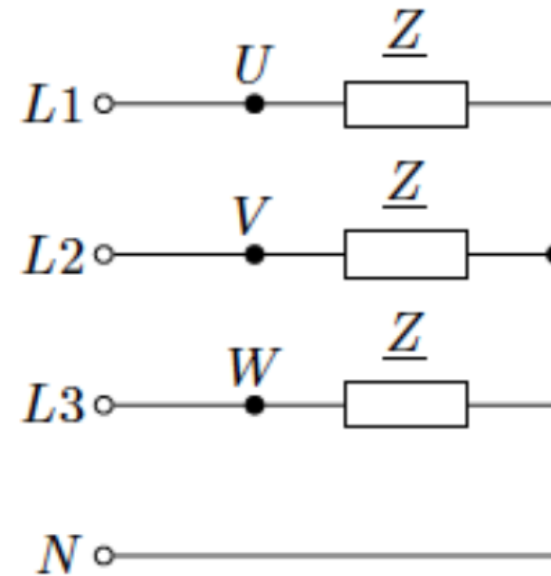
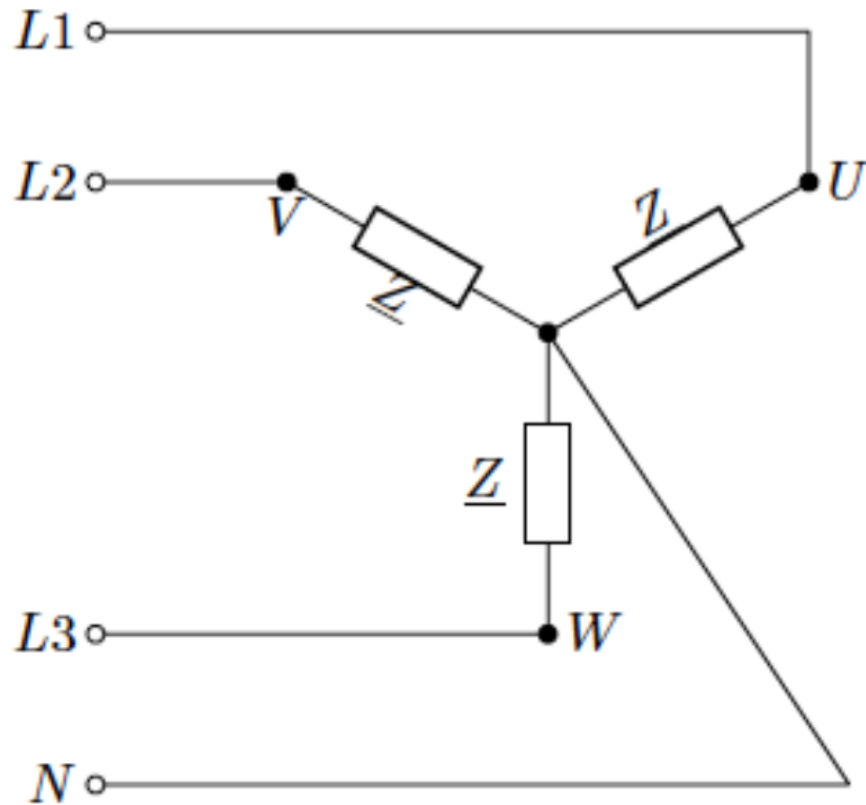
phase identifiers on motor terminals



Three-Phase Network



Star (Y) Connection



no current in N for symmetric loads (the three currents add to 0)
→ N could be removed

Phase Voltages

$$u_1 = \hat{u} \cdot \cos(\omega t)$$

$$u_2 = \hat{u} \cdot \cos\left(\omega t - \frac{2}{3}\pi\right)$$

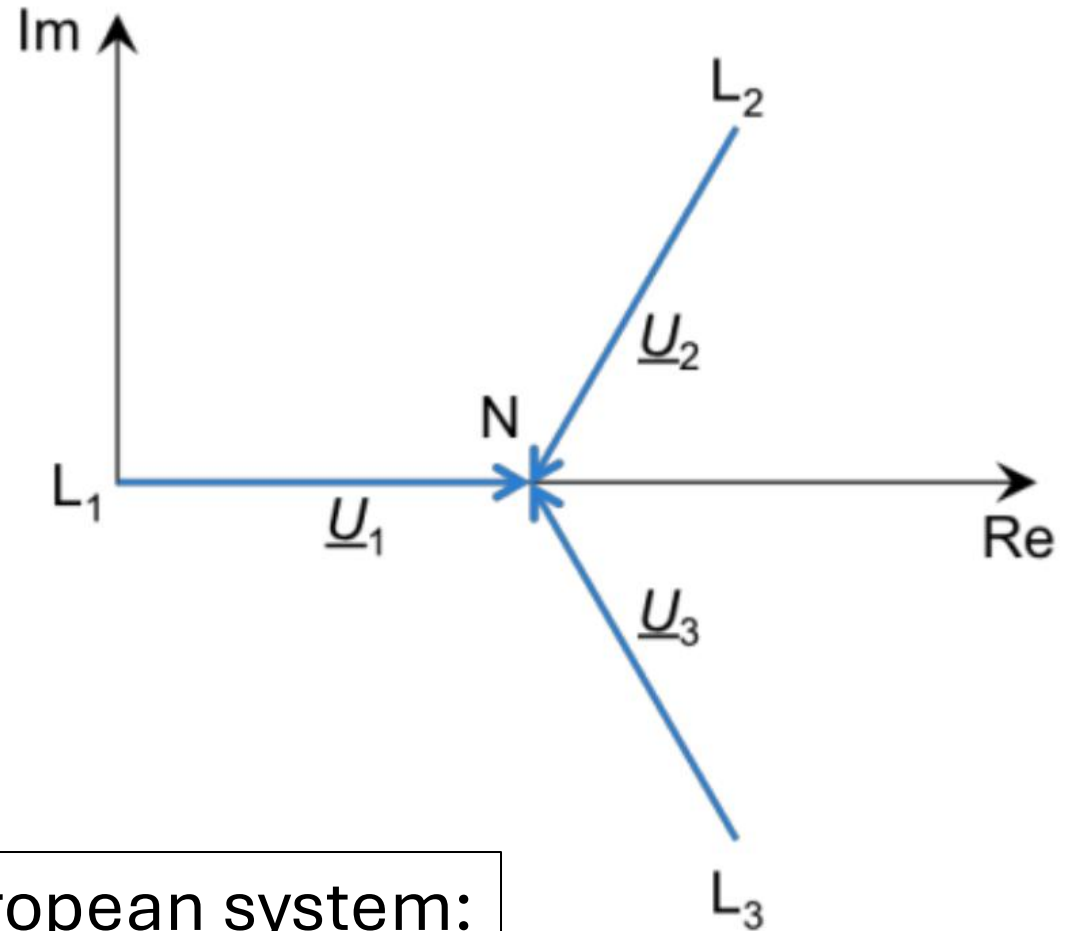
$$u_3 = \hat{u} \cdot \cos\left(\omega t - \frac{4}{3}\pi\right)$$

$$\underline{U}_1 = U$$

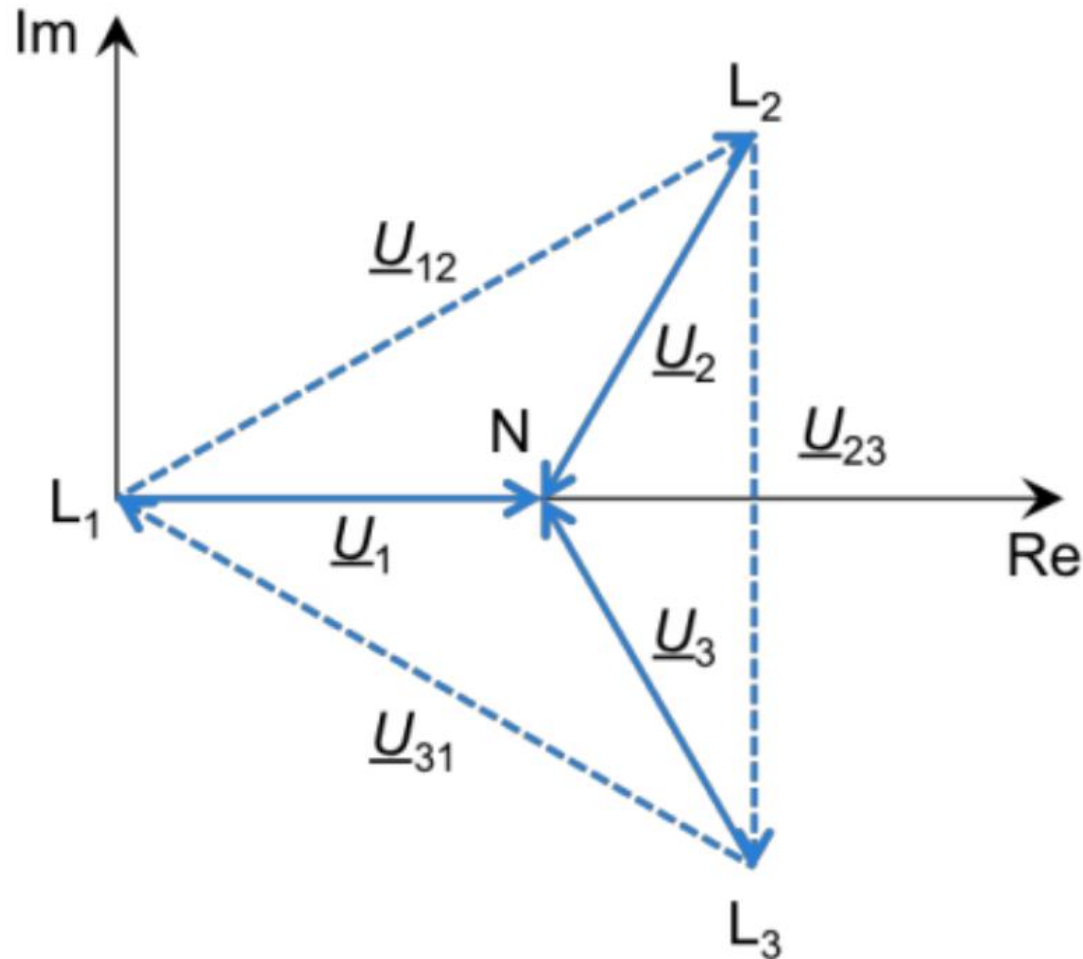
$$\underline{U}_2 = U \cdot e^{-j\frac{2}{3}\pi}$$

$$\underline{U}_3 = U \cdot e^{-j\frac{4}{3}\pi}$$

typical European system:
230 V



Line Voltages

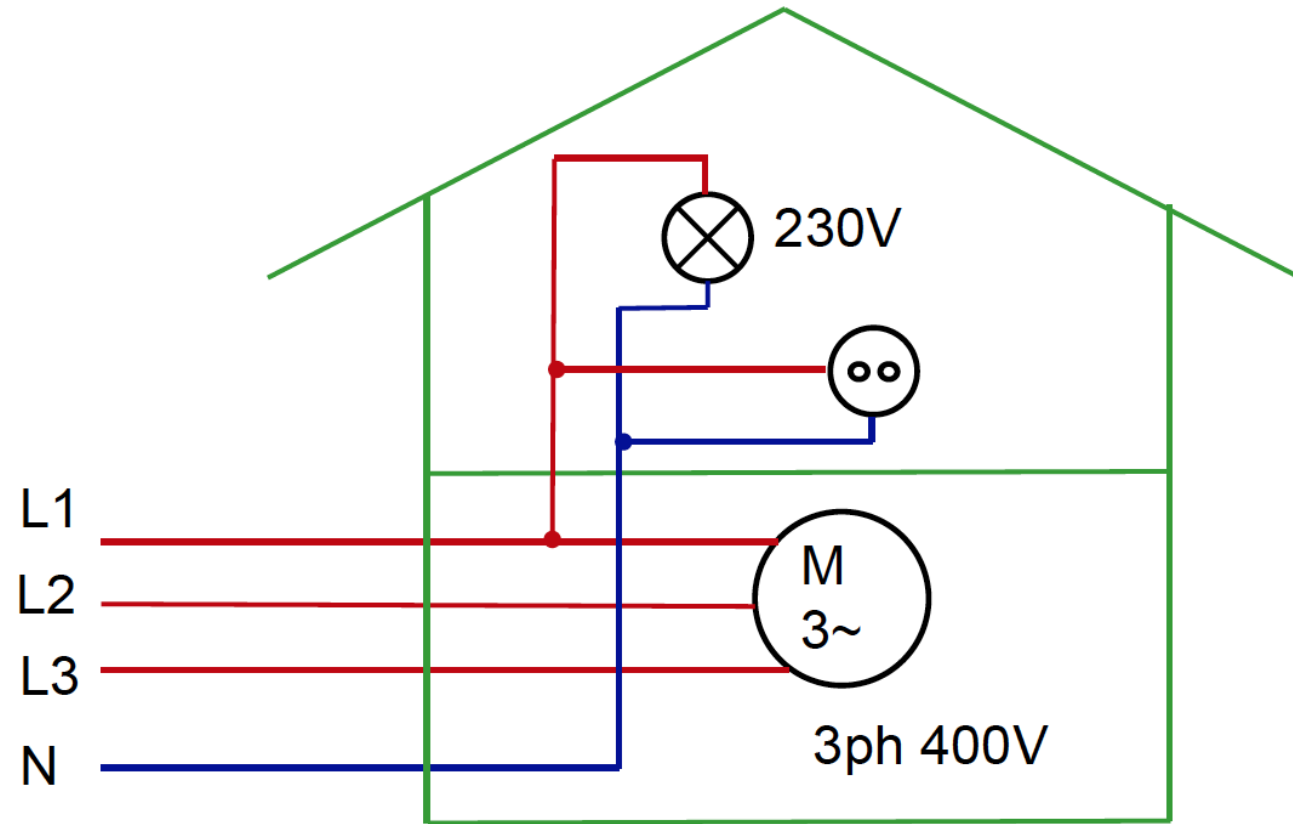


typical European system:
400 V

$$U_{12} = U_{23} = U_{31} = U_L = \sqrt{3} \cdot U$$

for example:

$$\begin{aligned} \underline{U}_{12} &= \underline{U}_1 - \underline{U}_2 = U - U \cdot e^{-j\frac{2}{3}\pi} = U \cdot \left(1 - \left(-0.5 - j\frac{\sqrt{3}}{2} \right) \right) \\ &= U \cdot 1.5 + j\frac{\sqrt{3}}{2}U = \sqrt{3} \cdot U \cdot e^{j\frac{\pi}{6}} \end{aligned}$$



Power for Symmetrical Load

$$\underline{I_1} = \underline{U_1} / \underline{Z}$$

$$\underline{I_2} = \underline{U_2} / \underline{Z}$$

$$\underline{I_3} = \underline{U_3} / \underline{Z}$$

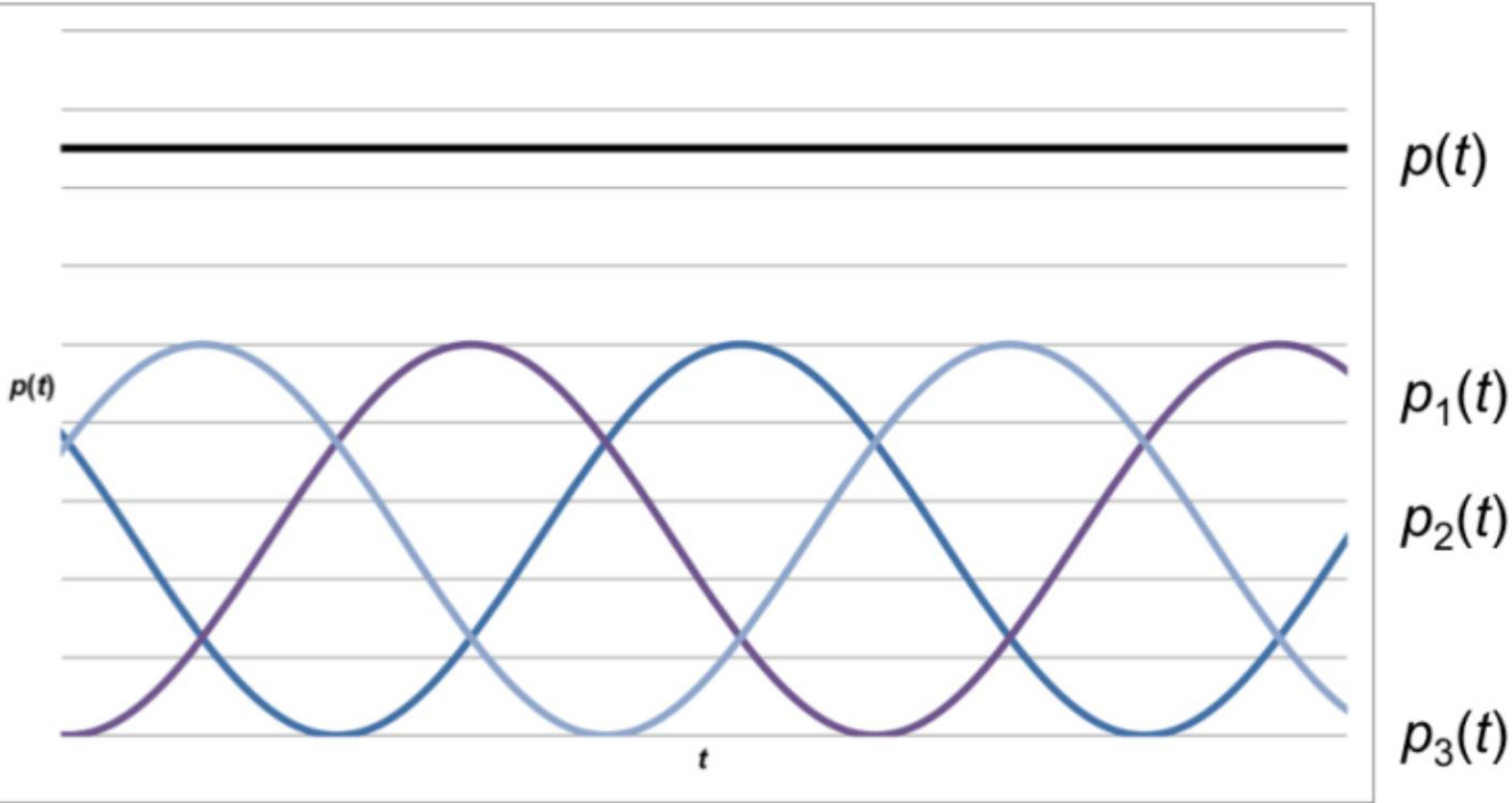
$$I_1 = I_2 = I_3 = I = I_L$$

$$P = 3 \cdot U_1 \cdot I_1 \cdot \operatorname{Re}(e^{j(\varphi_u - \varphi_i)}) = 3 \cdot U \cdot I \cdot \cos \varphi = \sqrt{3} \cdot U_L \cdot I_L \cdot \cos \varphi$$

$$Q = 3 \cdot U_1 \cdot I_1 \cdot \operatorname{Im}(e^{j(\varphi_u - \varphi_i)}) = \sqrt{3} \cdot U_L \cdot I_L \cdot \sin \varphi$$

$$S = \sqrt{P^2 + Q^2} = \sqrt{3} \cdot U_L \cdot I_L$$

Constant Power



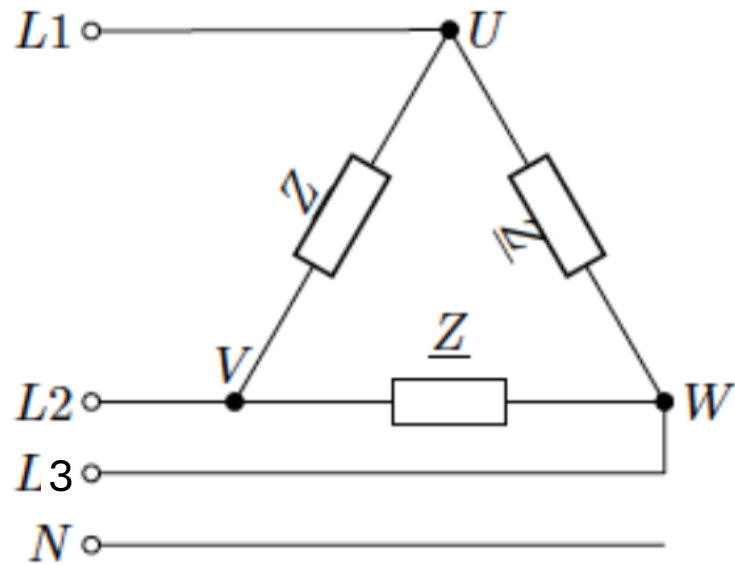
$$p(t) = \frac{u_1^2(t)}{R_1} + \frac{u_2^2(t)}{R_2} + \frac{u_3^2(t)}{R_3}$$

for symmetrical, purely resistive load:

$$R_1 = R_2 = R_3 = R$$

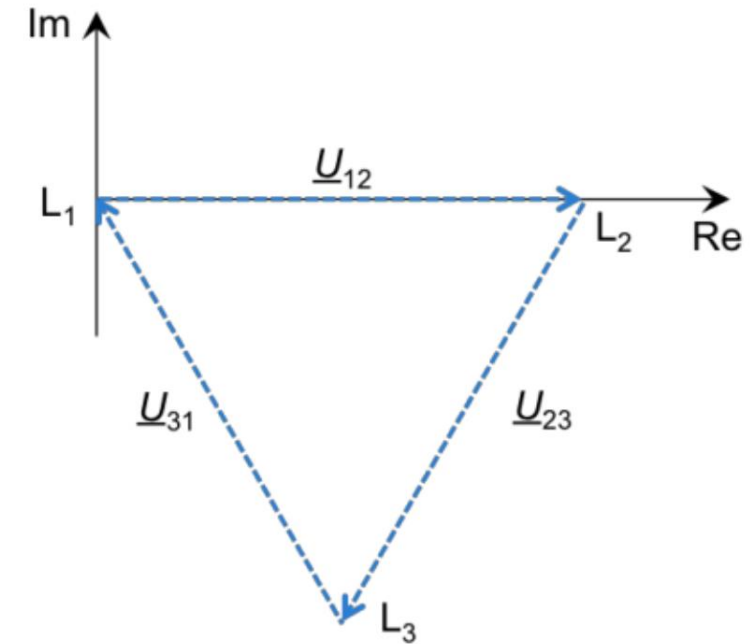
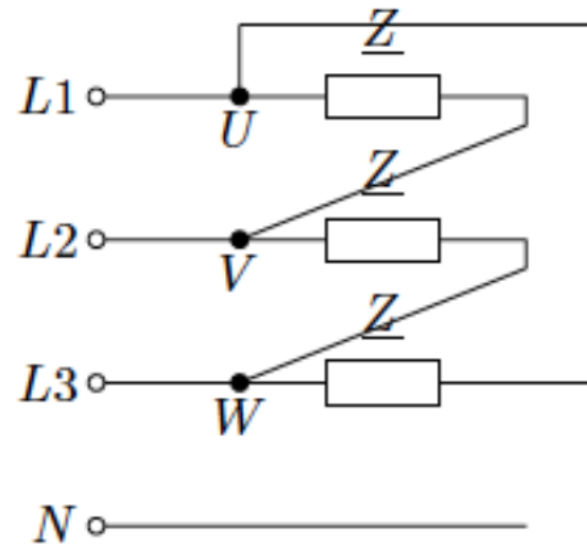
$$\rightarrow p(t) = \frac{\hat{U}^2}{R} \left(\cos^2 \omega t + \cos^2 \left(\omega t - \frac{2}{3} \pi \right) + \cos^2 \left(\omega t + \frac{2}{3} \pi \right) \right) = \frac{3\hat{U}^2}{2R} = \frac{3U^2}{R}$$

Delta (Δ) Connection



no neutral point

$$U_{12} = U_{23} = U_{31} = U_L = U = 400 \text{ V}$$



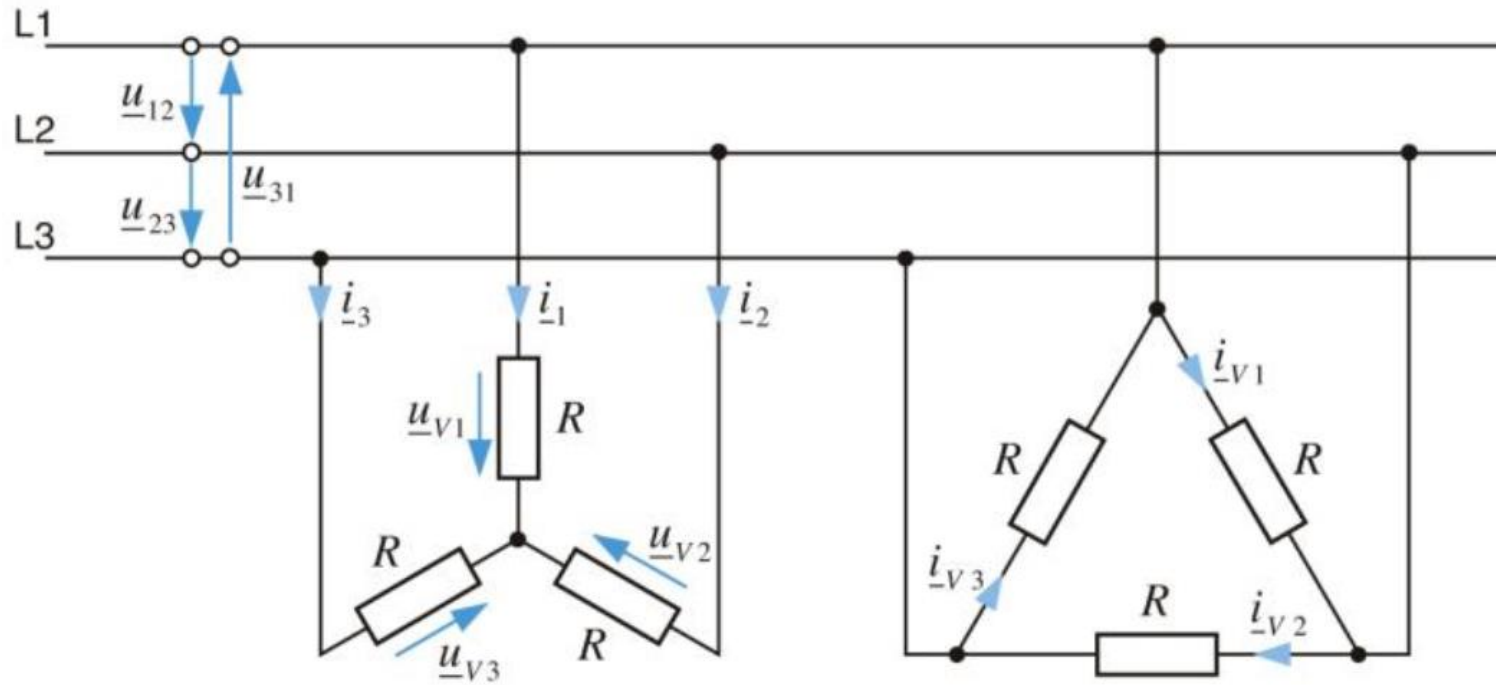
$$\underline{I}_{12} = \underline{U}_{12} / \underline{Z}$$

$$\underline{I}_{23} = \underline{U}_{23} / \underline{Z}$$

$$\underline{I}_{31} = \underline{U}_{31} / \underline{Z}$$

$$I_L = \sqrt{3} \cdot I$$

Power Consumption



for both star and delta:

$$P = \sqrt{3} \cdot U_L \cdot I_L \cdot \cos \varphi = 3 \cdot U \cdot I \cdot \cos \varphi$$

but for star:

$$U = U_L / \sqrt{3}, \quad I = U / R$$

$$\rightarrow P = \frac{U_L^2}{R} \cdot \cos \varphi$$

while for delta:

$$U = U_L, \quad I = U_L / R$$

$$\rightarrow P = 3 \cdot \frac{U_L^2}{R} \cdot \cos \varphi$$

$$\rightarrow P_{\Delta} = 3 \cdot P_Y$$